

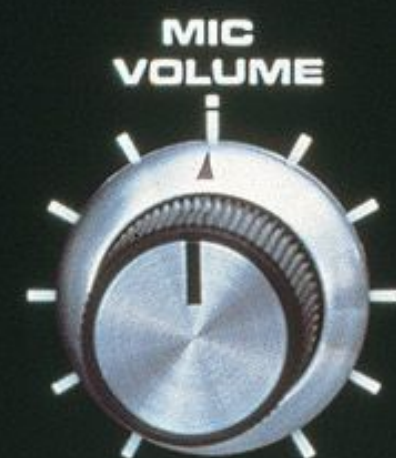
AUDIO/VIDEO TECH VL03



SPACE ECHO RE-201



PEAK
LEVEL



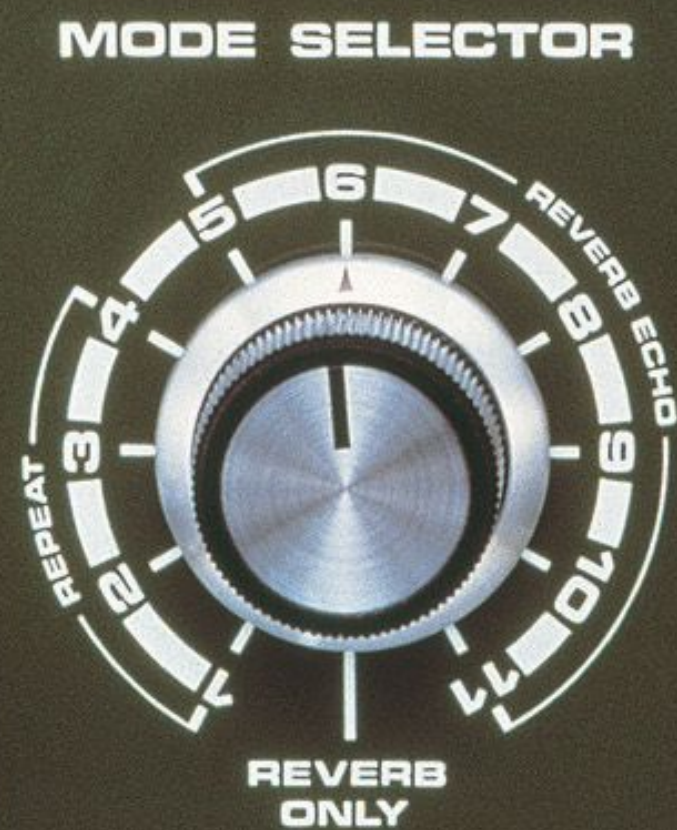
MIC
VOLUME



MIC
VOLUME



INSTRUMENT
VOLUME



MODE SELECTOR

REVERB
ONLY



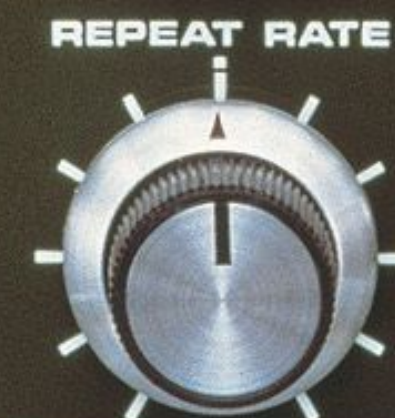
BASS



TREBLE



REVERB VOLUME



REPEAT RATE



INTENSITY



ECHO VOLUME

STRAIGHT ECHO



ON
OFF



POWER

MIC



MIC



INSTRUMENT



ECHO NORMAL

FROM P.A.



OUT PUT

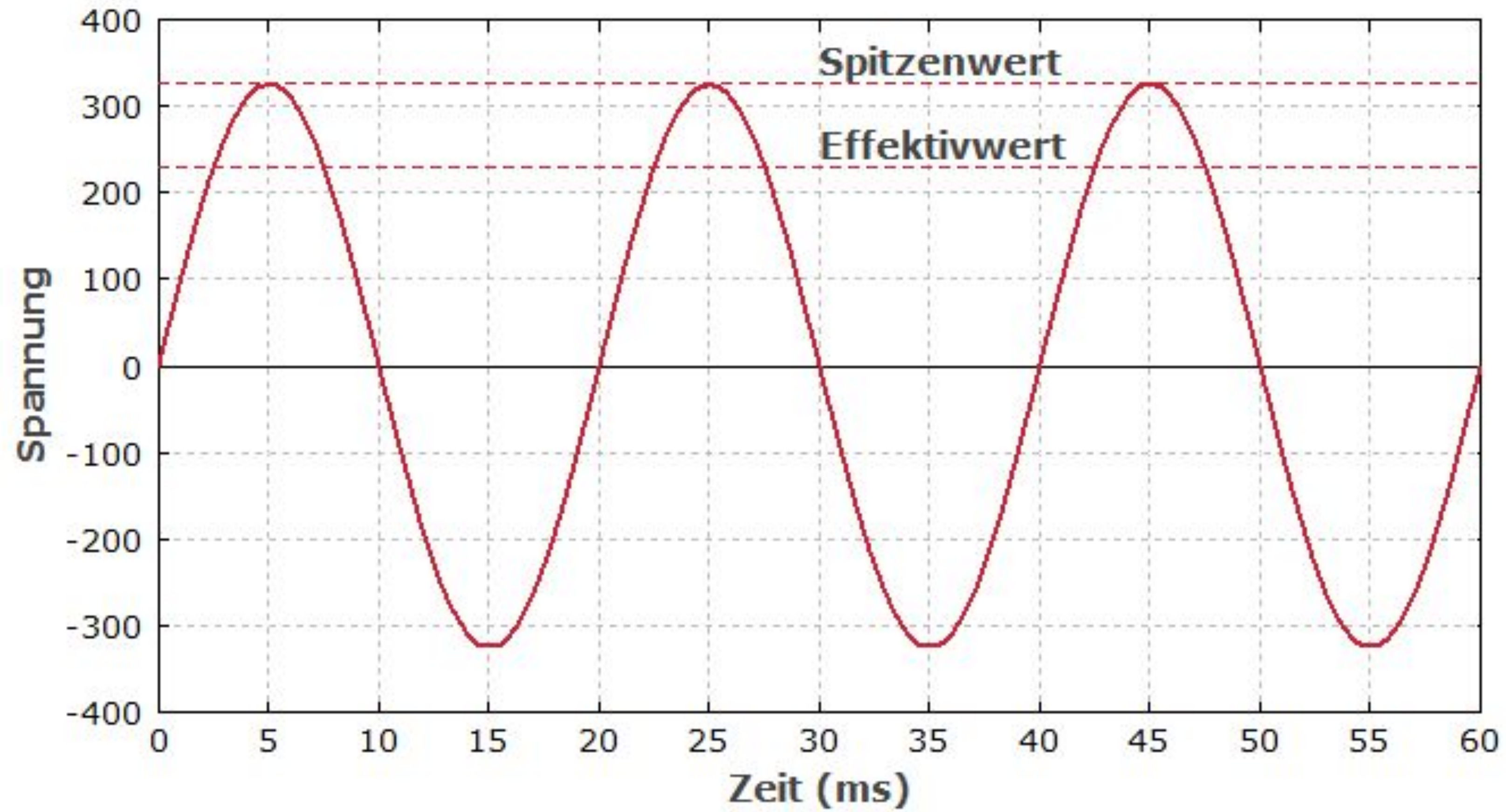


FOOT SW



ECHO CANCEL

WIEDERHOLUNG

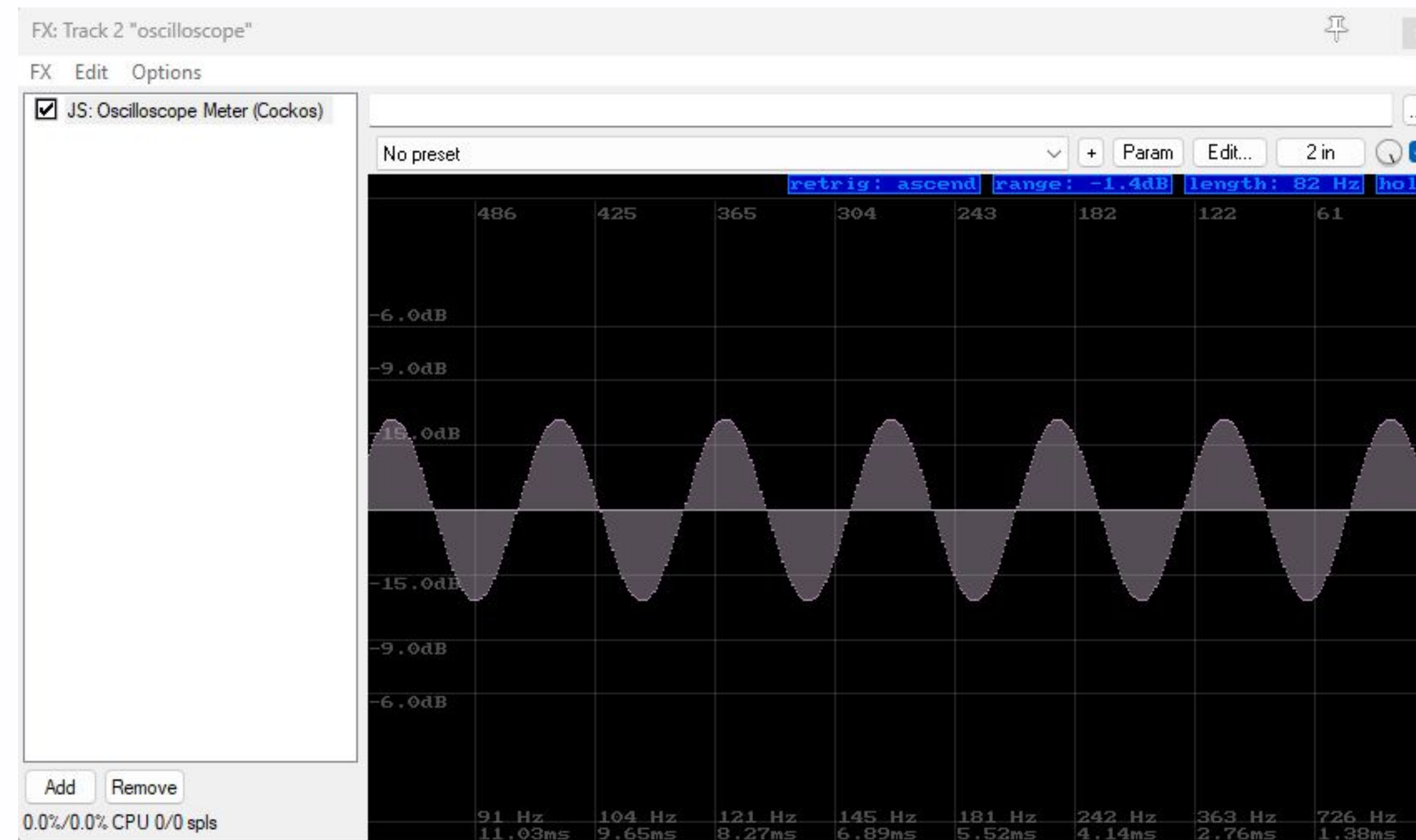


Wechselspannung

periodische Schwingung

Schwingungsdauer T, zB. 20ms

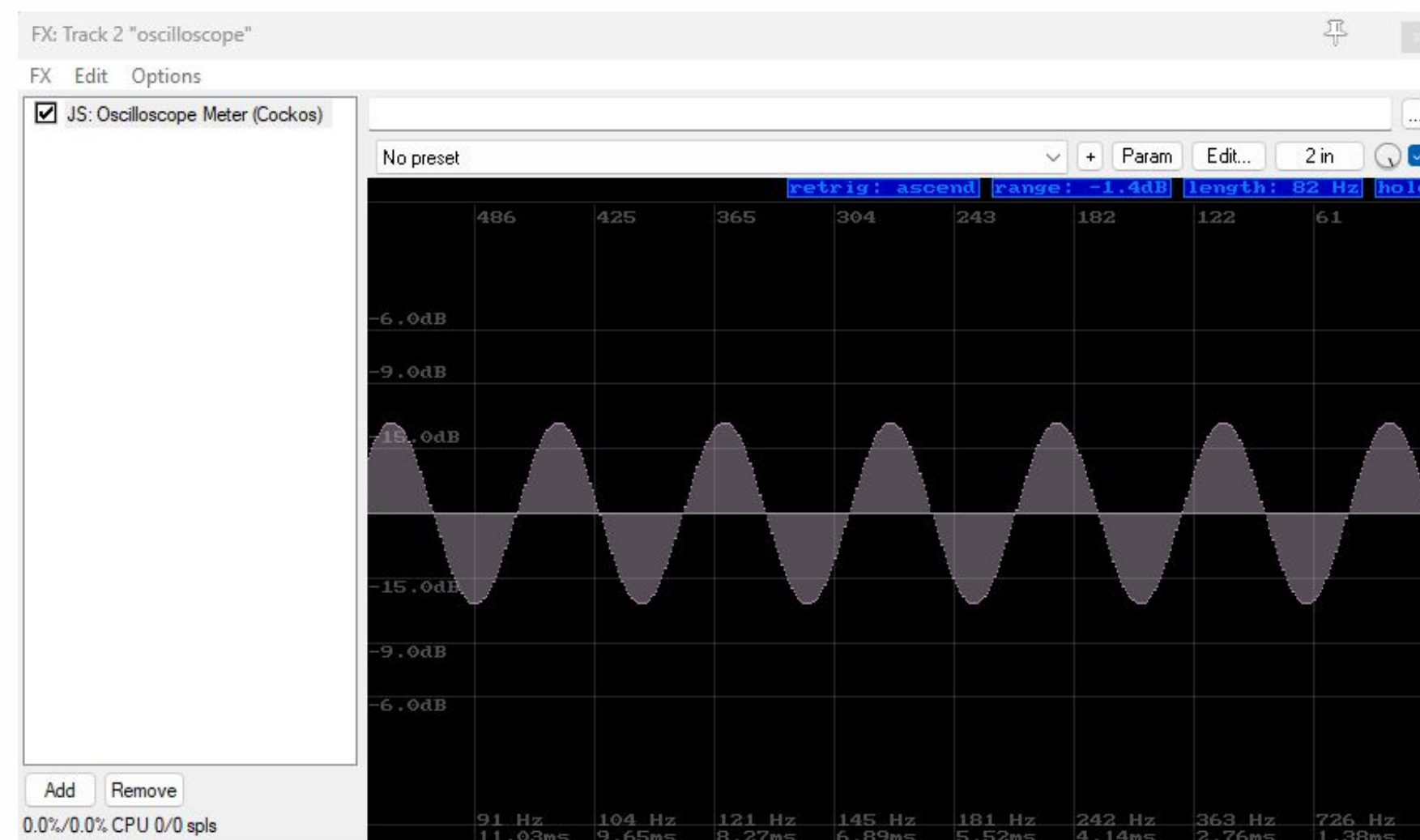
Frequenz $f = 1 / T = 1 / 0,02 = 50\text{Hz}$



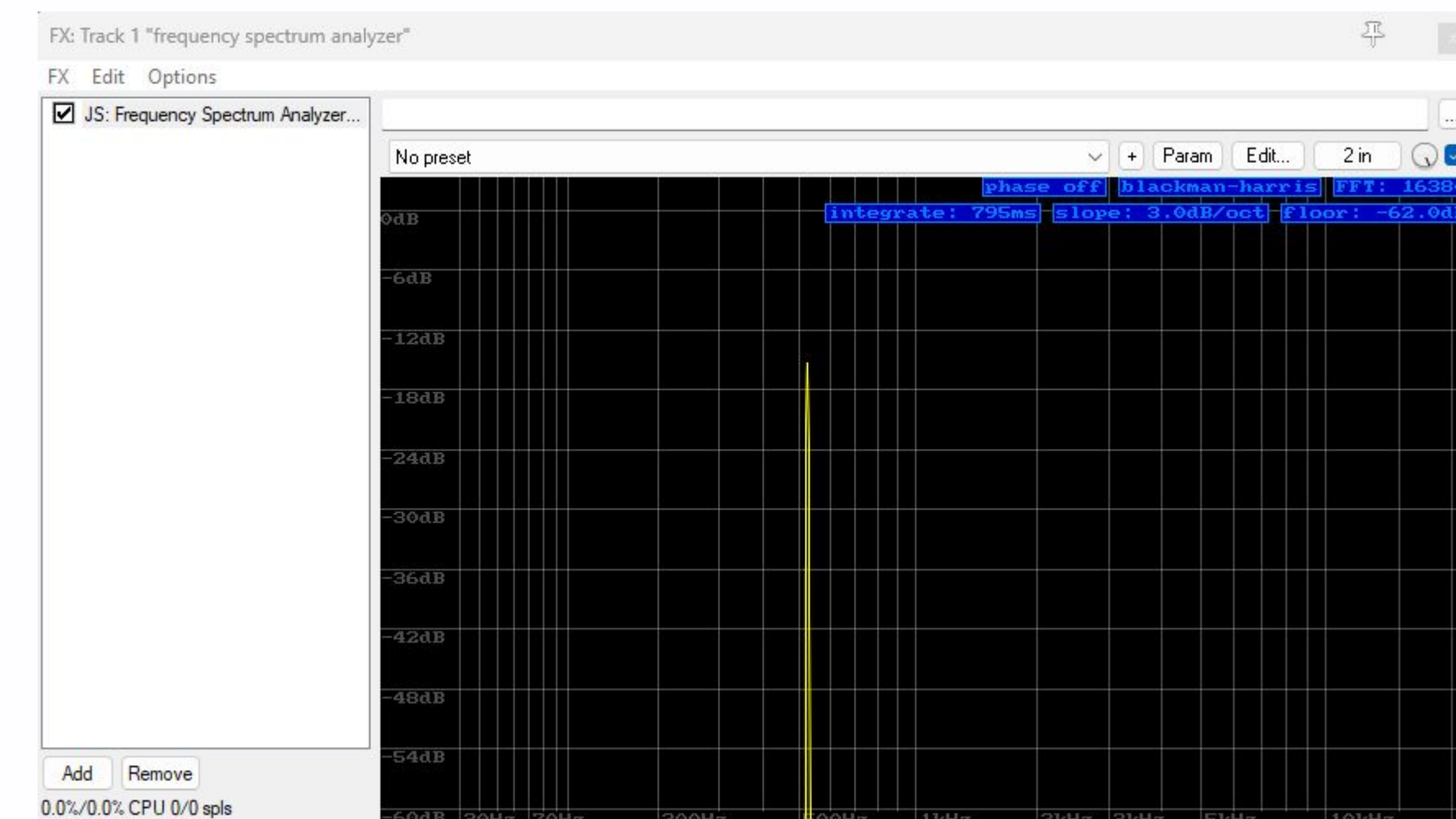
Audiosignal Sinus

Fourierreihe: Jede periodische Funktion $f(t)$ mit Schwingungsdauer T kann in Sinus- und Cosinus-Schwingungen zerlegt werden!

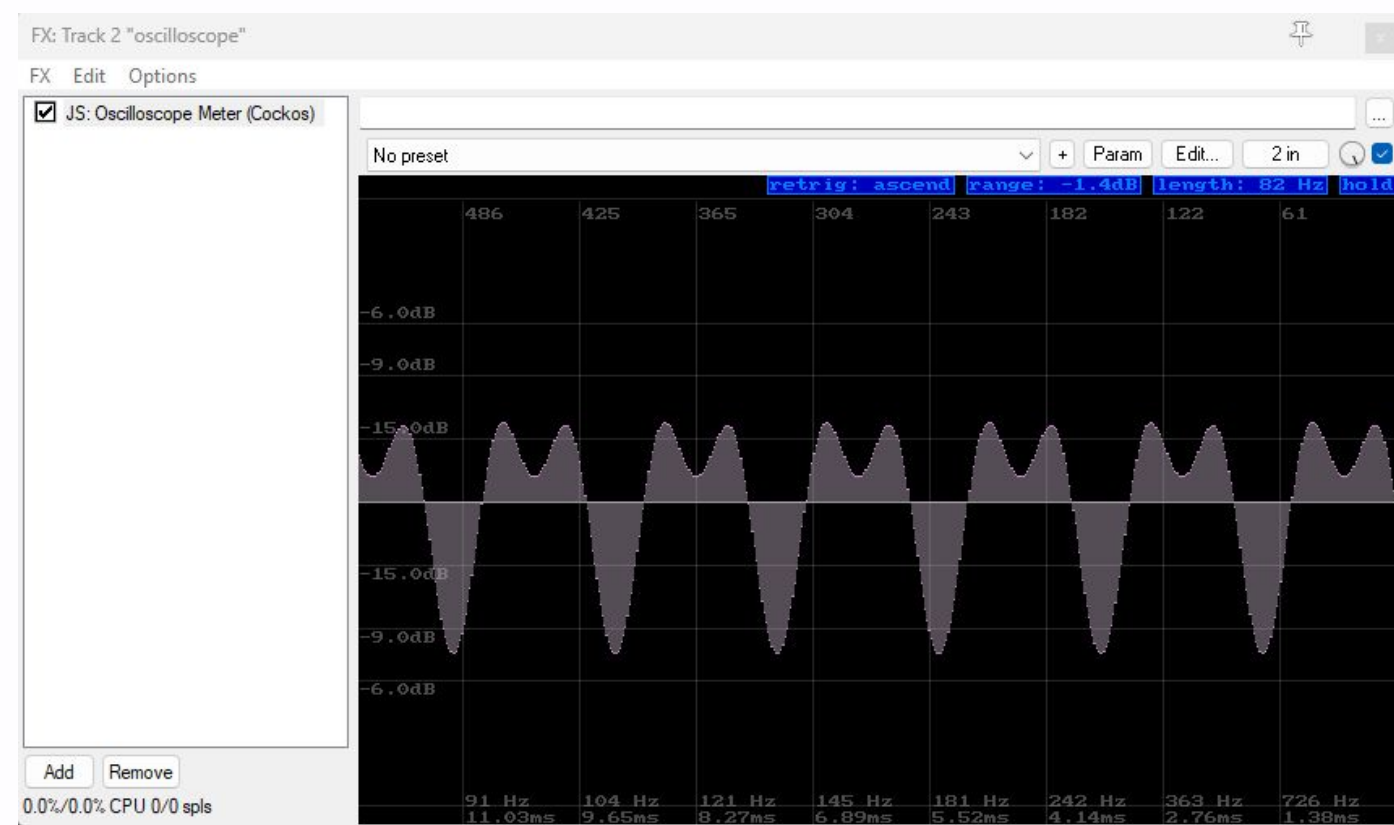
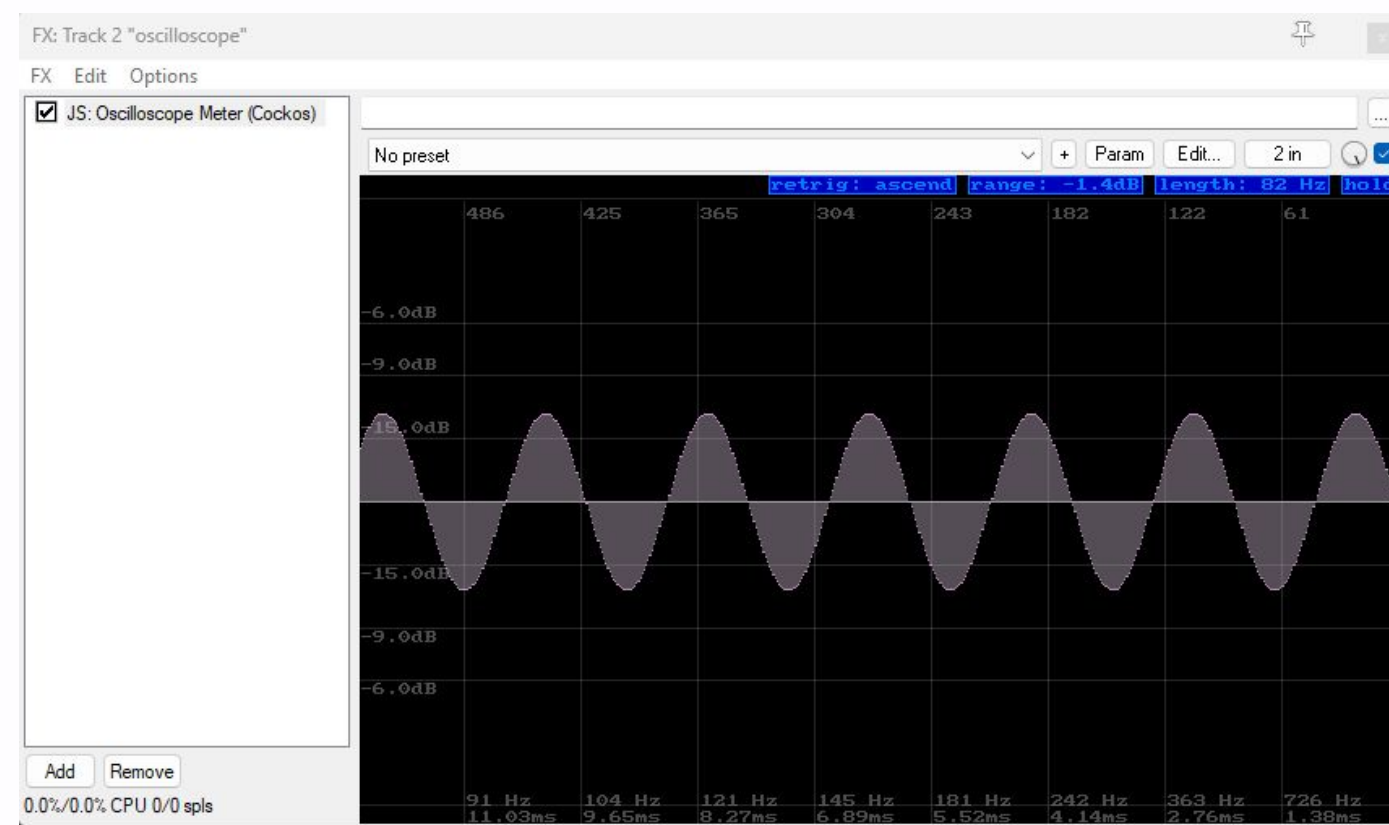
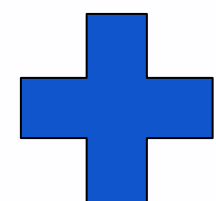
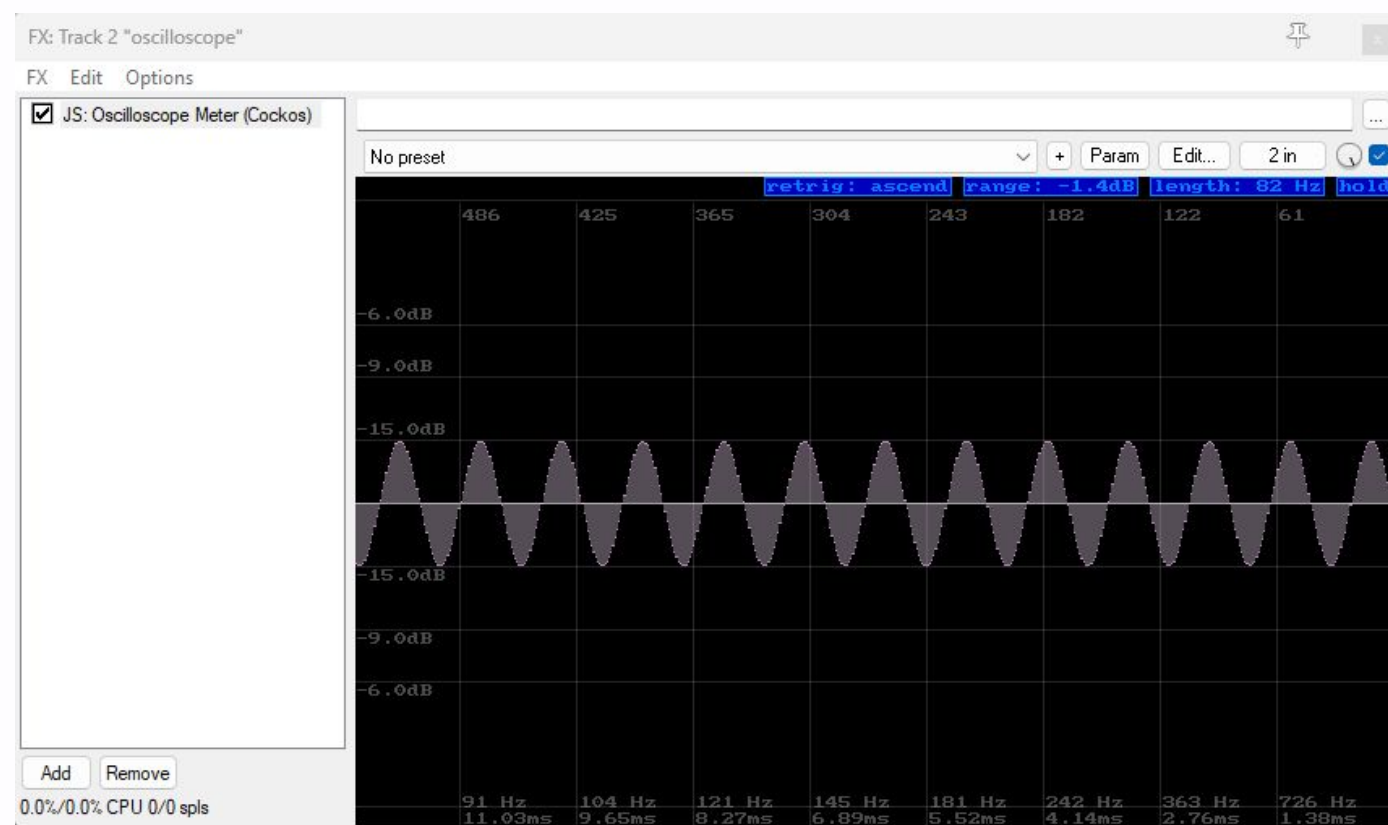
Die Frequenzen der Teilschwingungen sind dabei ganzzahlige Vielfache der Grundfrequenz.



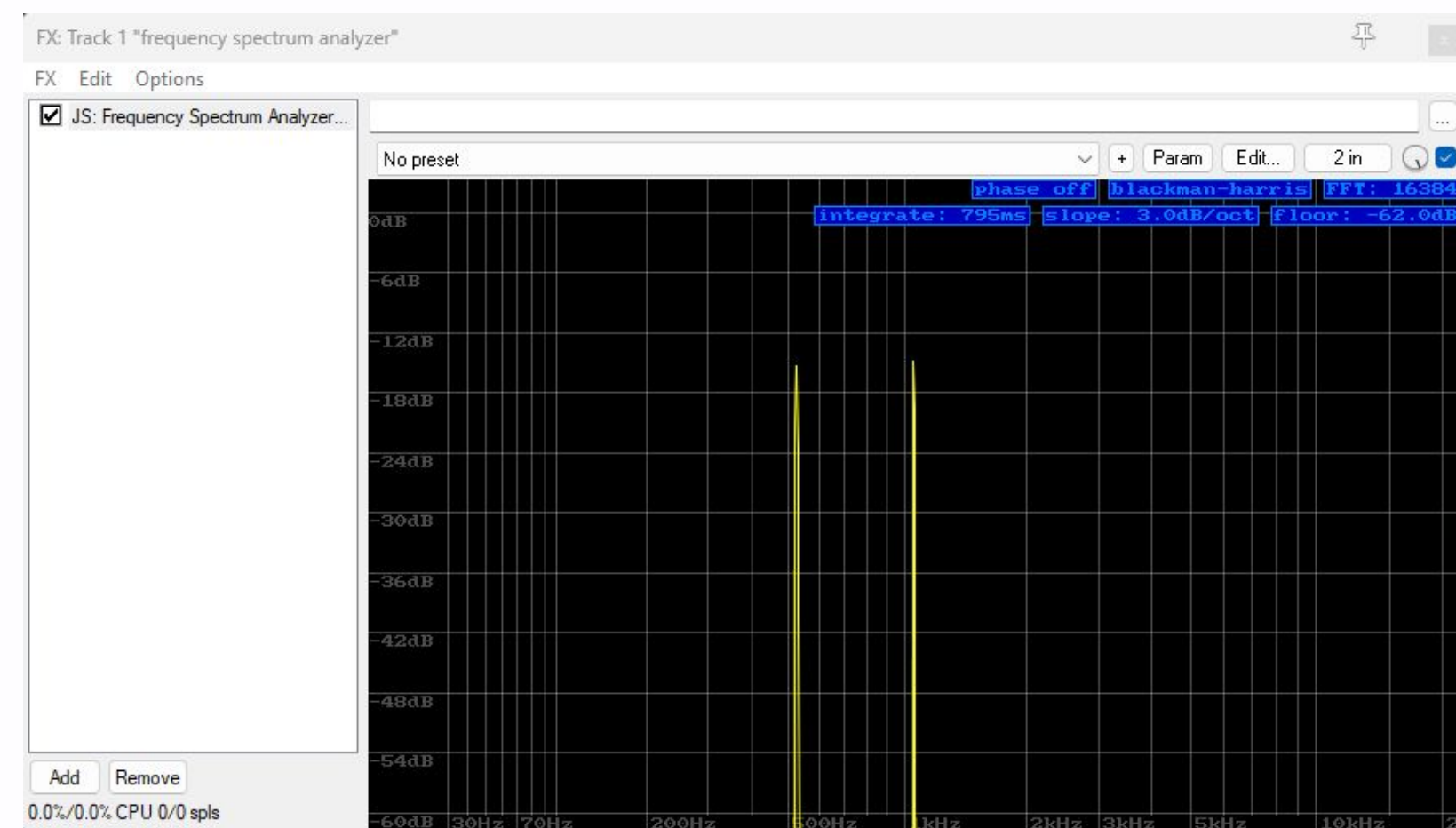
Fourier Analyse



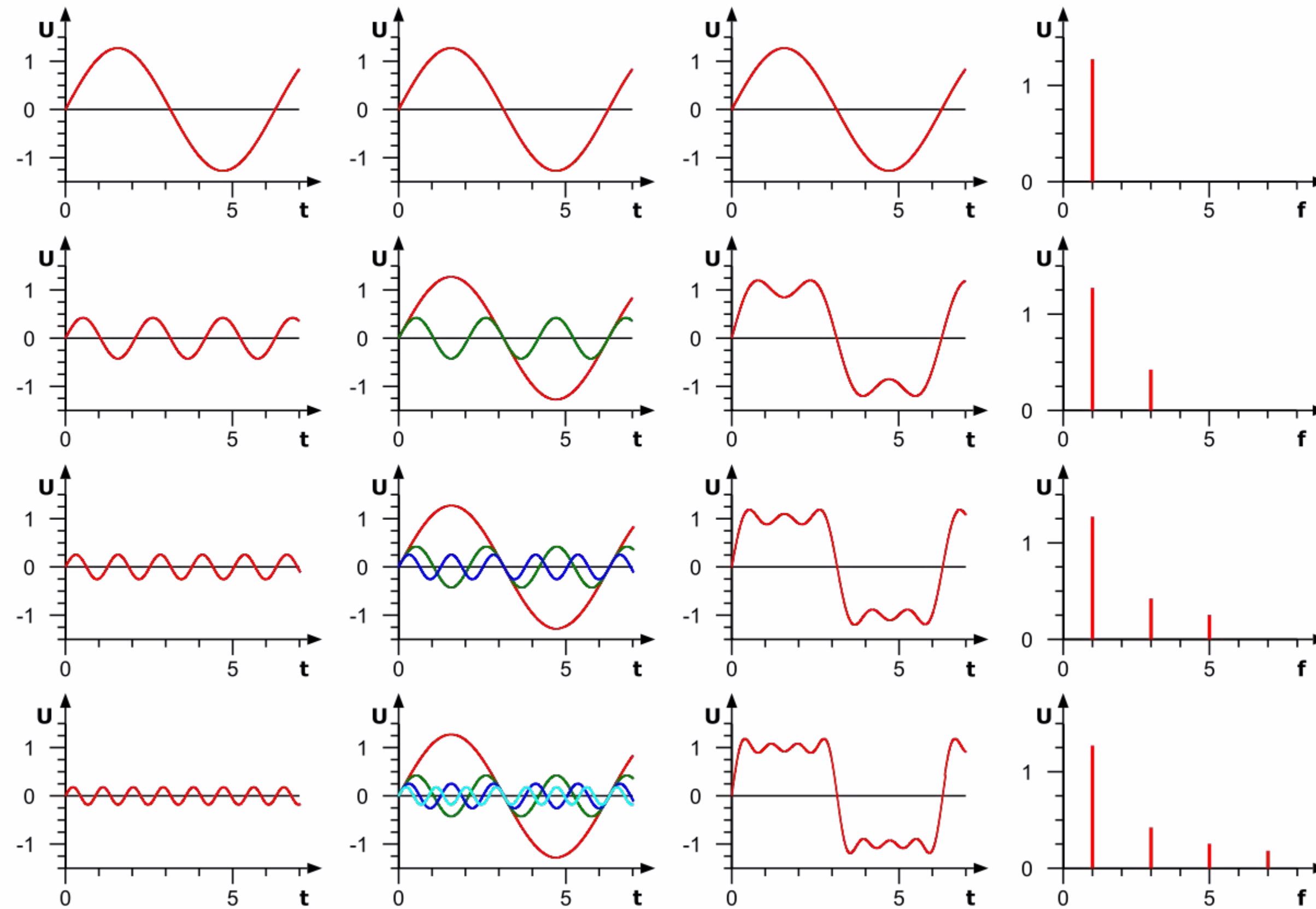
Audiosignal Sinus



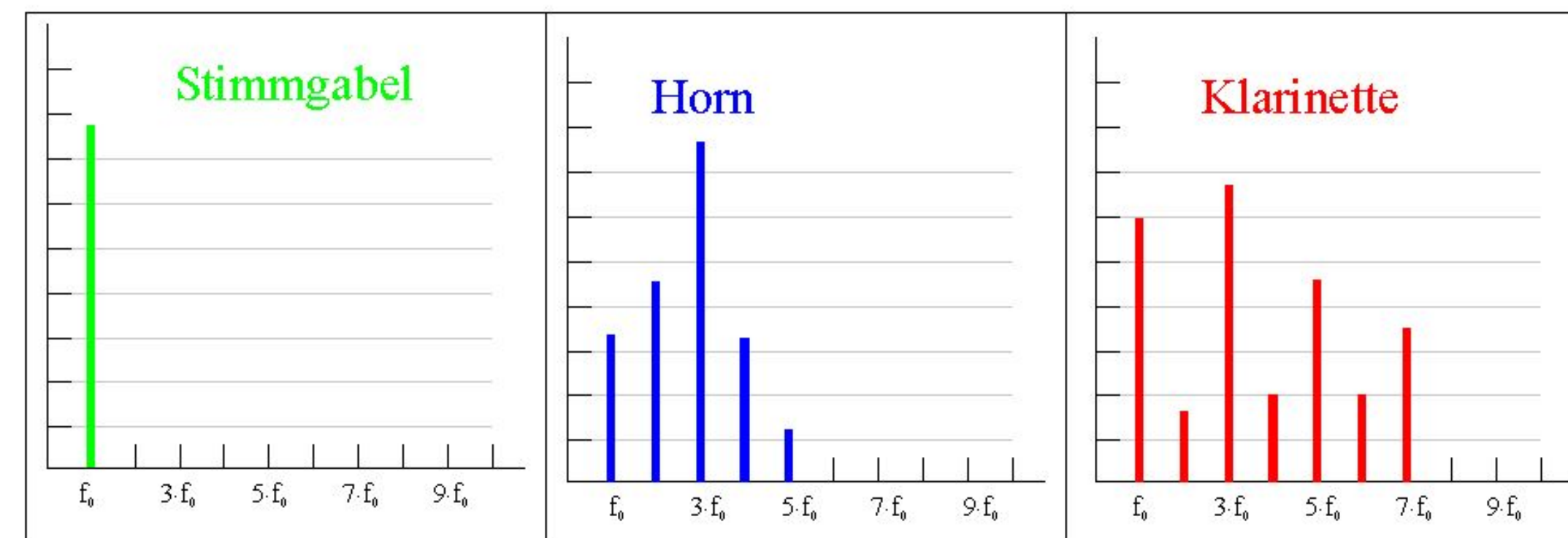
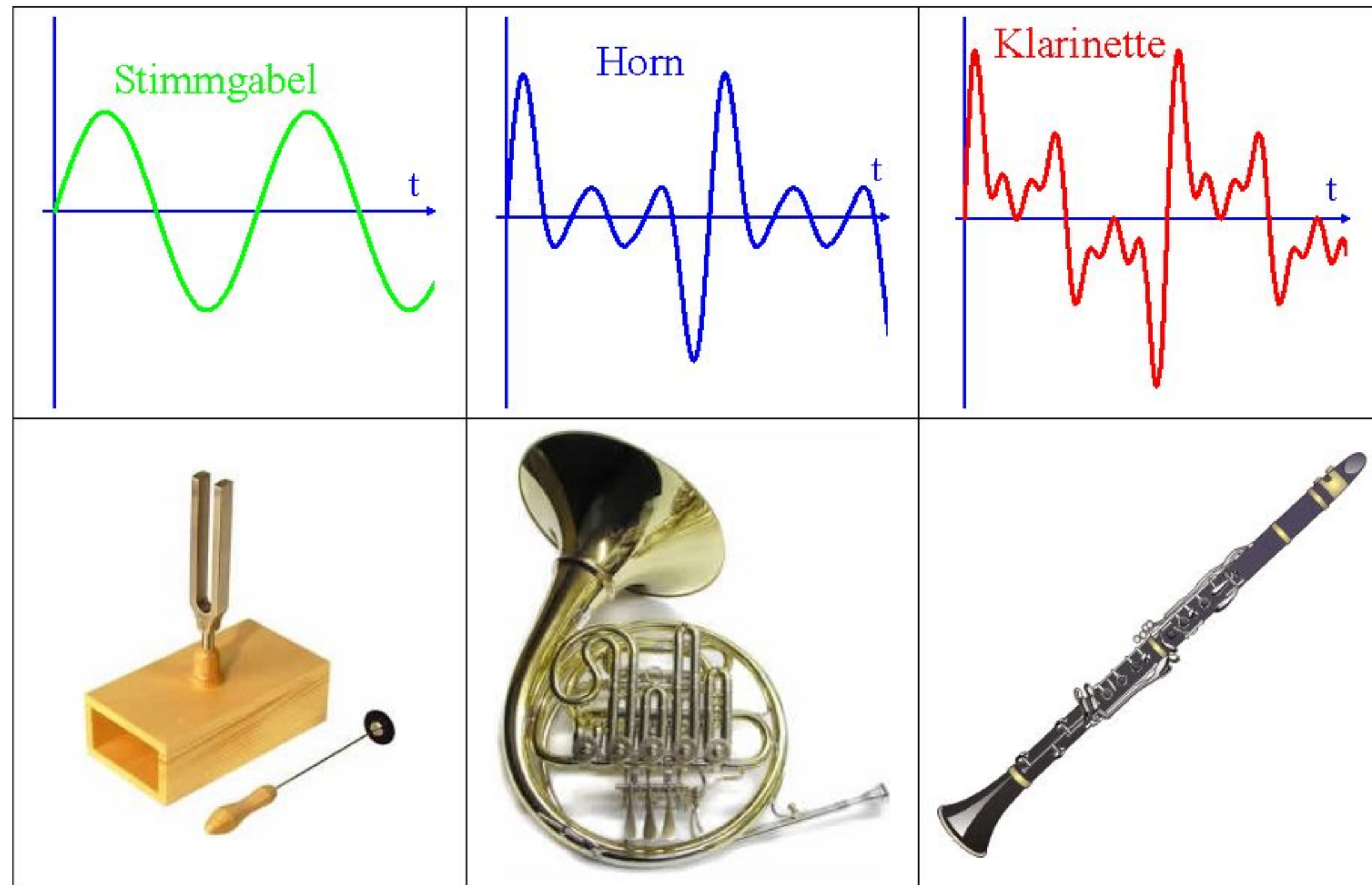
Fourier Analyse



Überlagerung von 2 Schwing



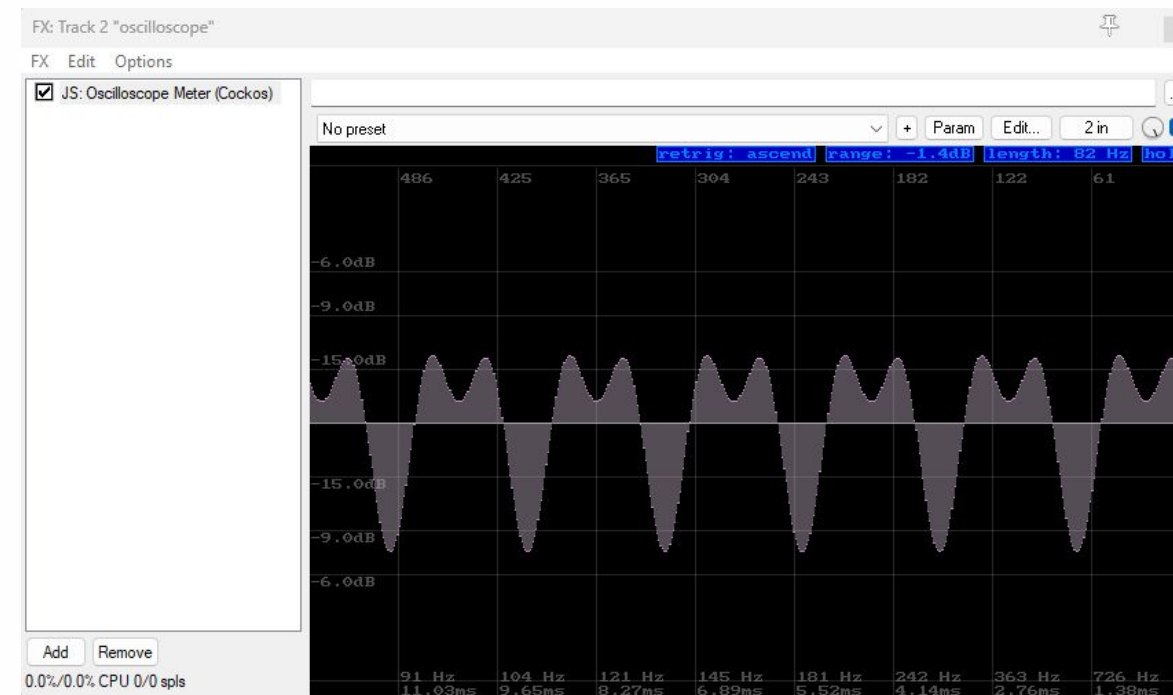
Umkehrung: Fourier-Synthes
einer Rechtecksfunktion



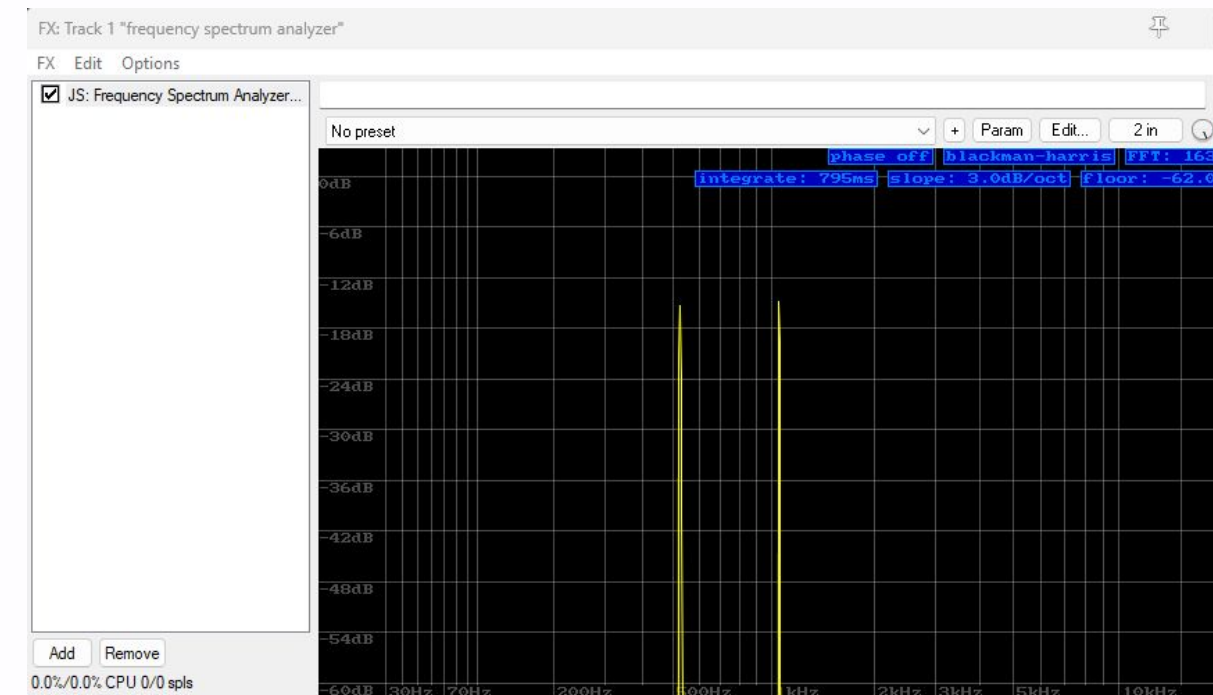
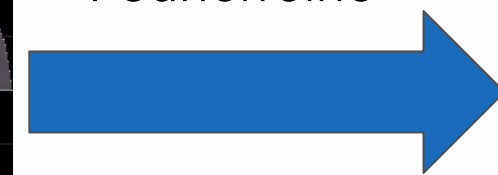
Obertonreihe, Klangfarbe

time domain

frequency domain

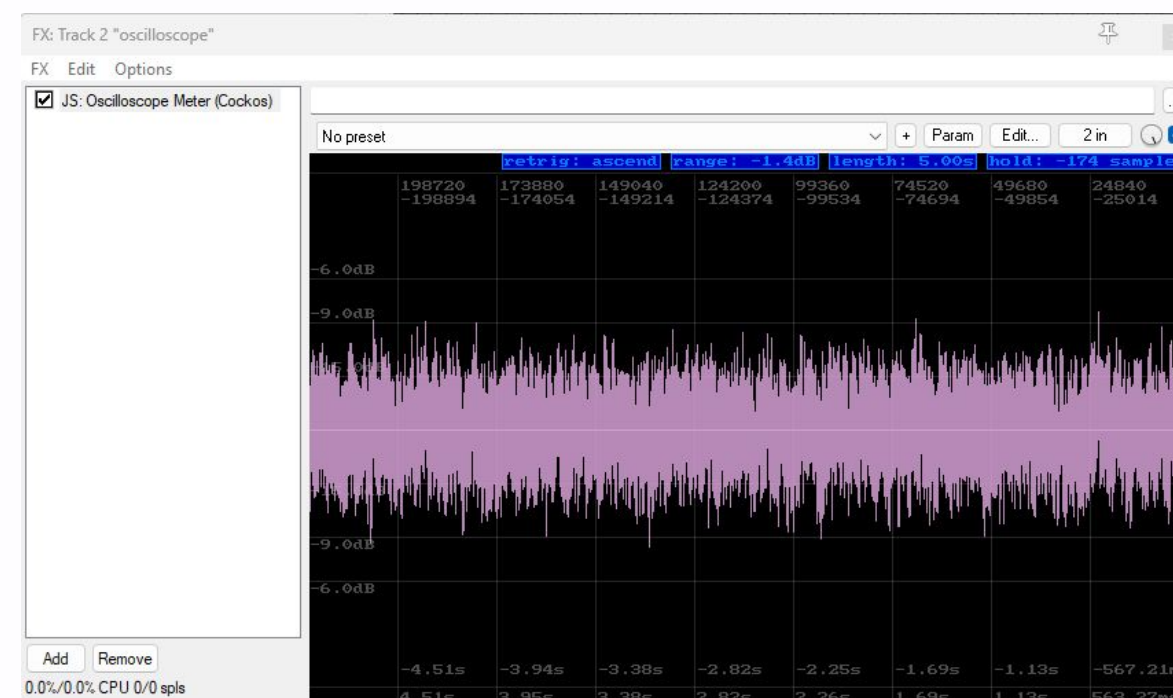


Fourierreihe



periodisches Signal: Übersetzung über Fourierreihe in Grundschiwingung und Obertöne

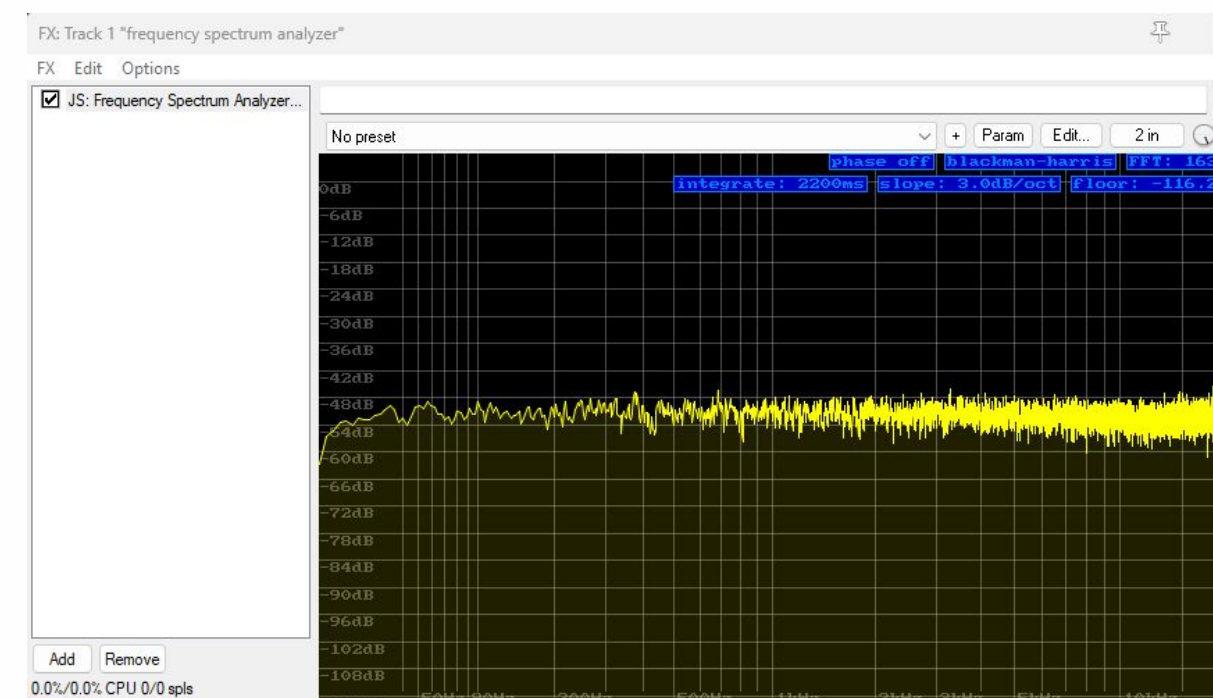
A



Fourier-Transformation



A

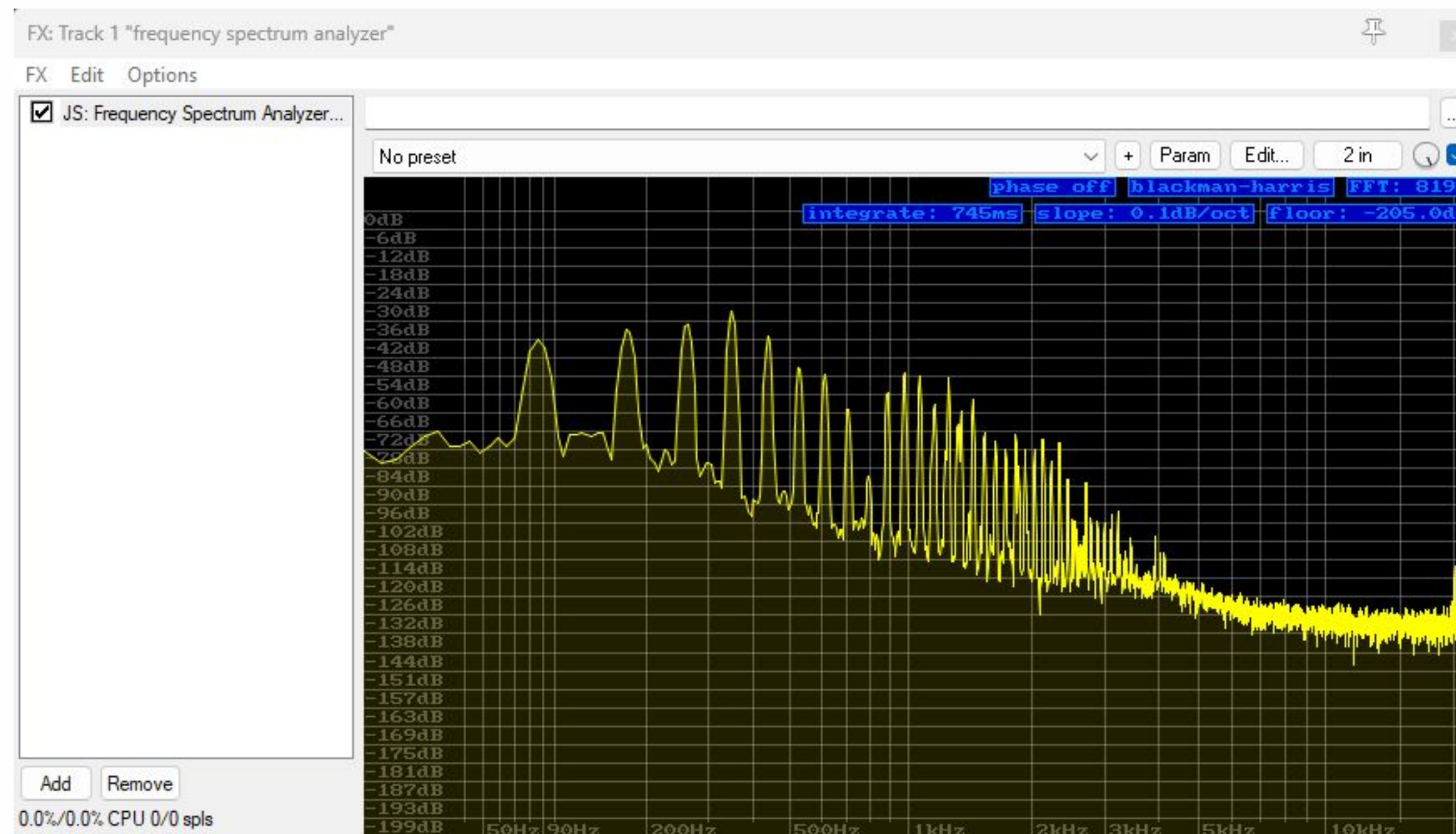


nicht-periodisches Signal: Übersetzung über Fouriertransformation in ein kontinuierliches Spektrum
Zeitfenster wird analysiert (hier: Rauschen)

t

f

diskretes vs kontinuierliches Frequenzspektrum



Mischung aus periodischen und nicht-periodischen Anteilen

zB Snaredrum: viel Rauschanteil

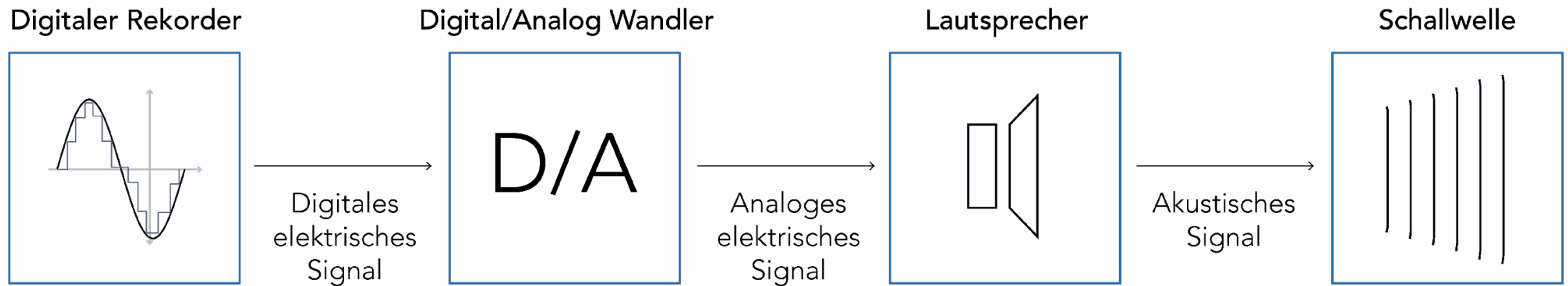
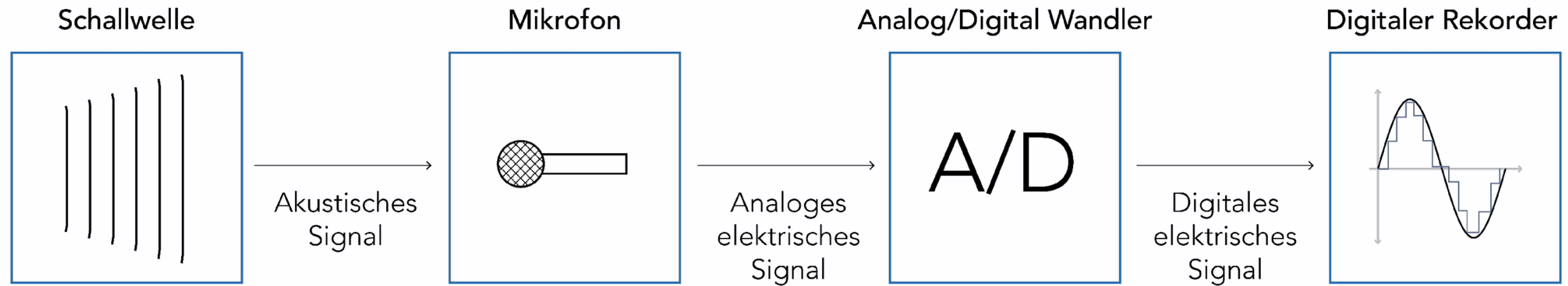
AUDIOSIGNAL



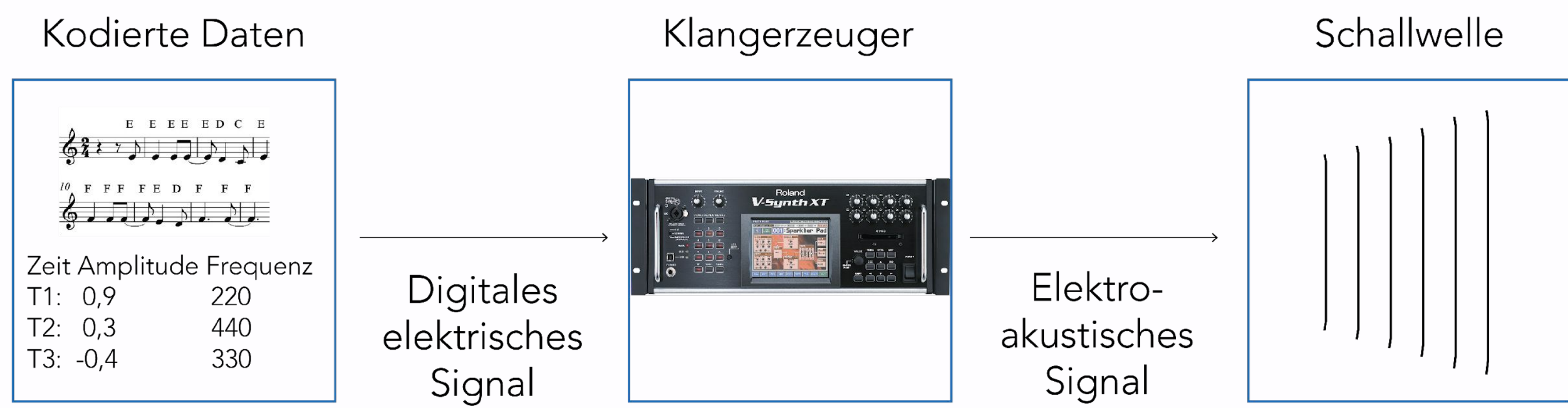
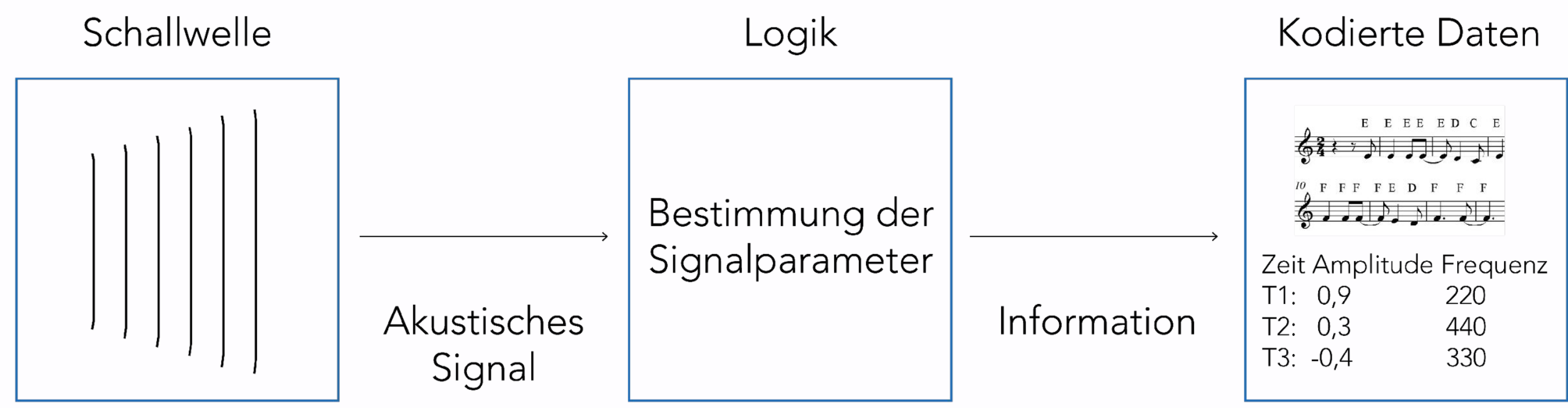
SPEICHERUNG
AUDIOSIGNAL

The screenshot displays the Ableton Live software interface, specifically the Region and Track sections for a track named 'Inst 2'. The Region section at the top includes controls for Mute, Loop, Quantize (set to 'off'), Q-Swing, Transpose, and Velocity. A 'More' button is also present. The Track section for 'Inst 2' is divided into two main areas. The left area contains a 'Session B...' button, a waveform display, a 'MIDI FX' button, and a stack of effect buttons: 'EXS24', 'Compressor', 'Bass Amp', and 'Channel EQ'. Below these are 'Send' and 'Stereo Out' buttons, a 'Read' button, a volume knob set to 3.7, and a vertical gain slider ranging from 0 to 60. The right area contains a 'Setting' button, an 'EQ' button, a 'Limiter' button, another 'Read' button, a volume knob set to -5.2, and a vertical gain slider ranging from 0 to 60. At the bottom, the track is labeled 'Inst 2' and 'Output 1-2'.

The screenshot displays a digital audio workstation (DAW) interface. The top section shows a waveform view for a track named "04 The Vacuum". The waveform is blue and shows a complex, oscillating signal. Below the waveform, there are tabs for "Piano Roll", "Score", and "Step Editor". The "Piano Roll" tab is selected, showing a musical score with 11 measures. The score is written in bass clef, 4/4 time, and features various musical notes and rests. A blue box with white text is overlaid on the bottom right of the piano roll, reading "SIGNALFORM VS PARAMETRISCH".



**SIGNALFORM
KODIERUNG**



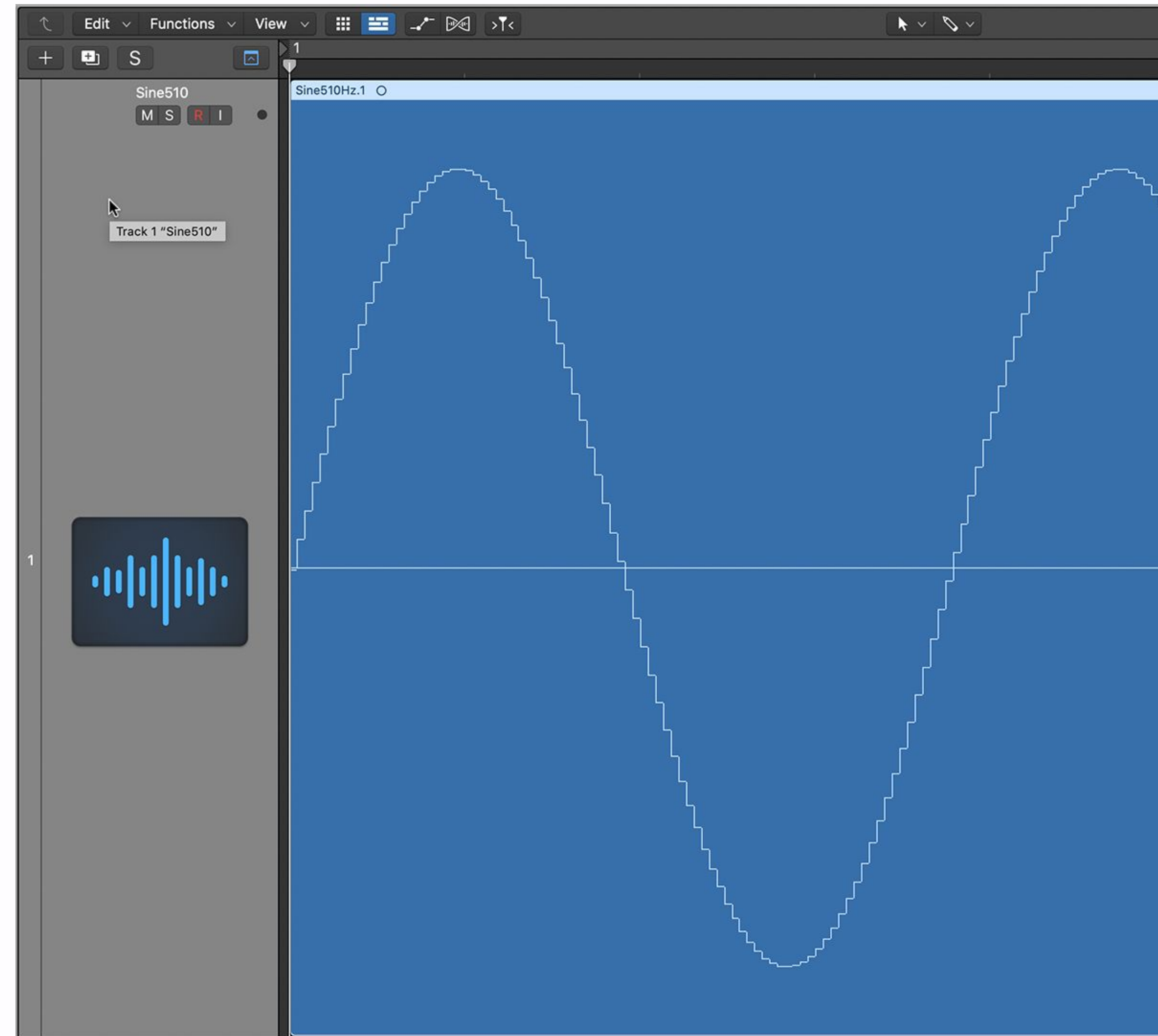
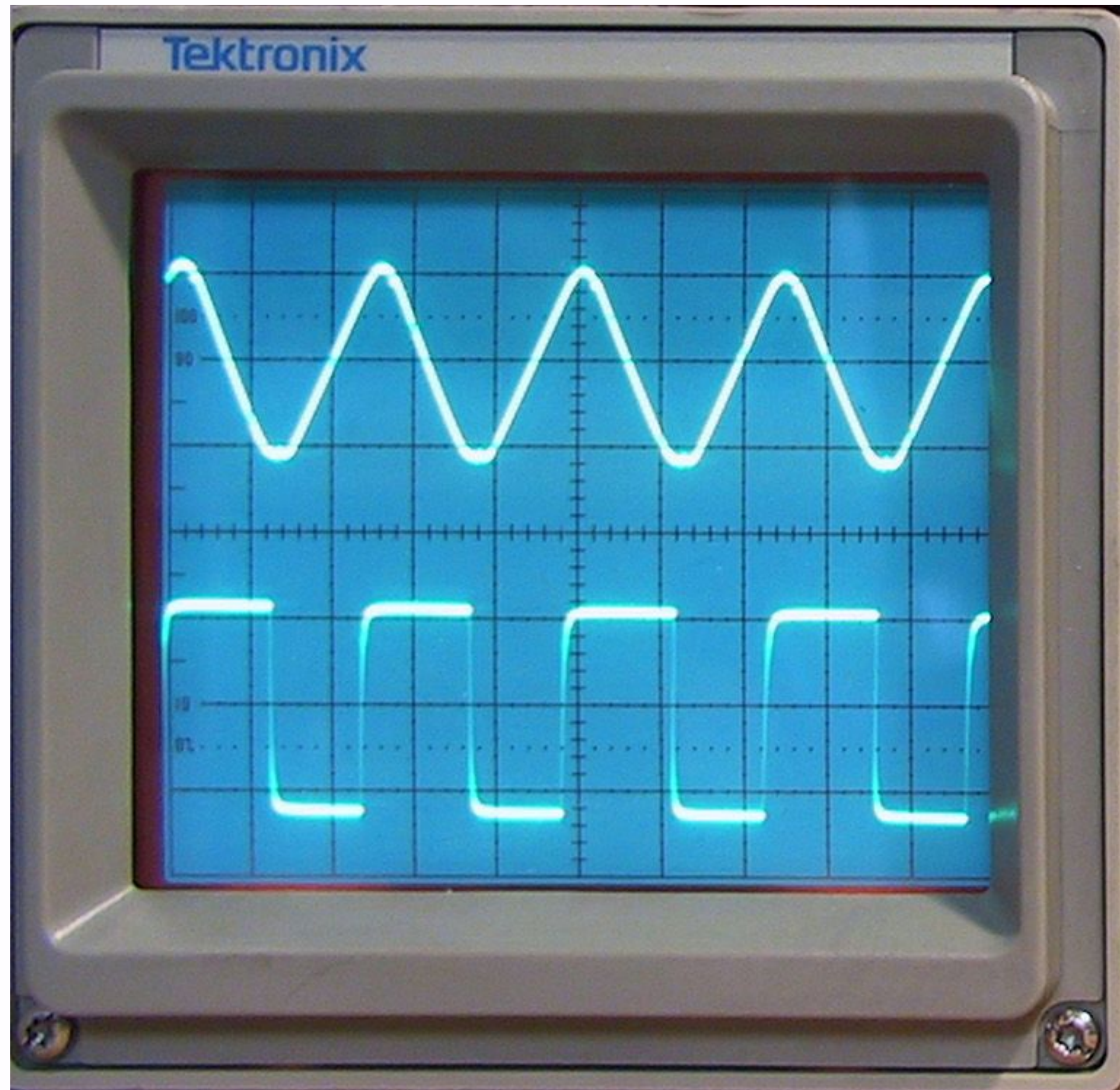
PARAMETRISCHE
KODIERUNG

KODIERUNG	SIGNALFORM	PARAMETRISCH
FILEGRÖSSE	DAUER	KOMPLEXITÄT
SPEICHERBEDARF	HOCH	GERING
KLANGERZEUGUNG	NICHT NOTWENDIG	NOTWENDIG
REPRODUKTION	ORIGINALGETREU	ABWEICHEND
WIEDERGABE	STARR	ANPASSBAR
FORMATE	PCM (AIFF/WAV), CD, LP...	MIDI, NOTATION, MPEG7...

EIGENSCHAFTEN ÜBERSICHT



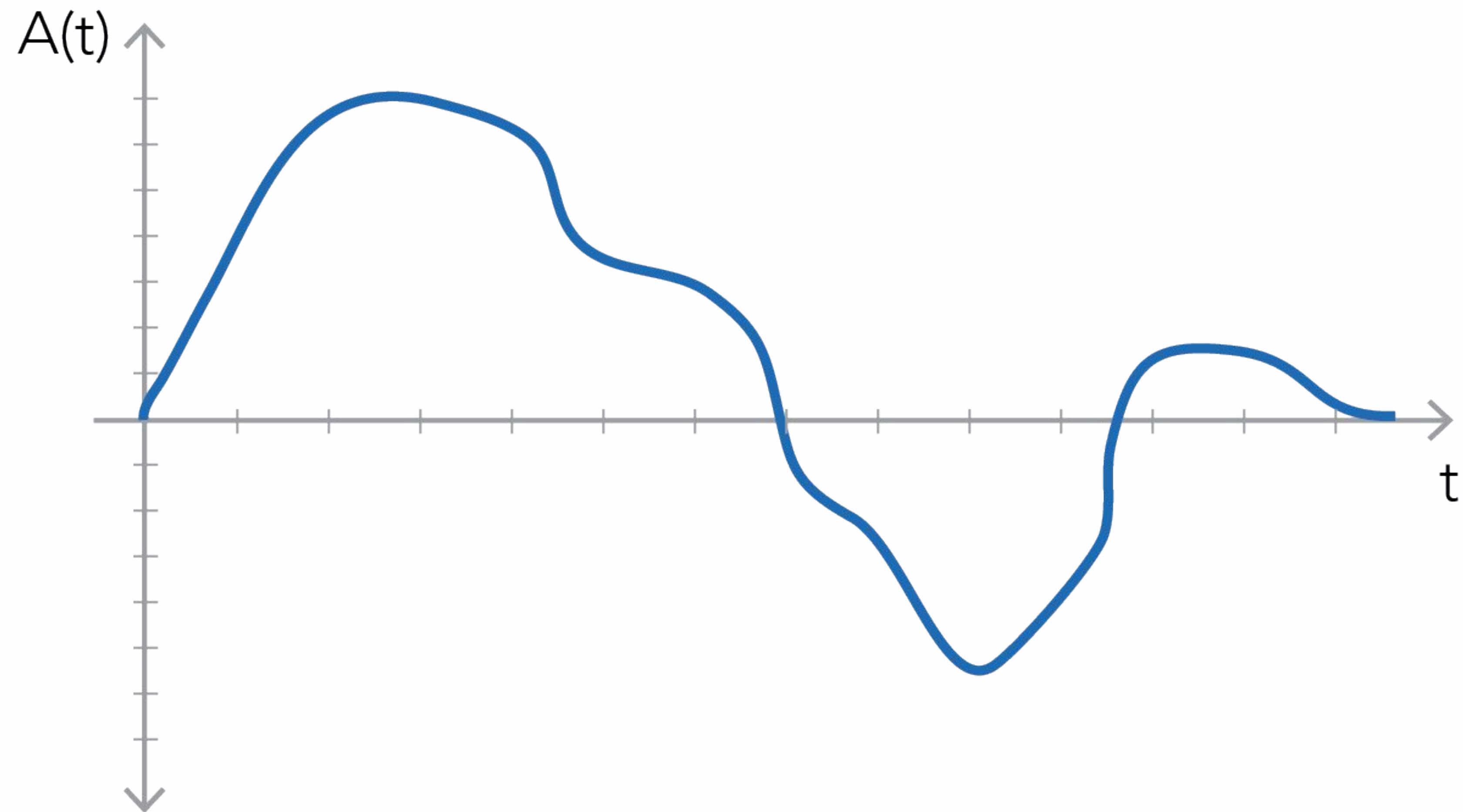
DIGITALSYSTEM



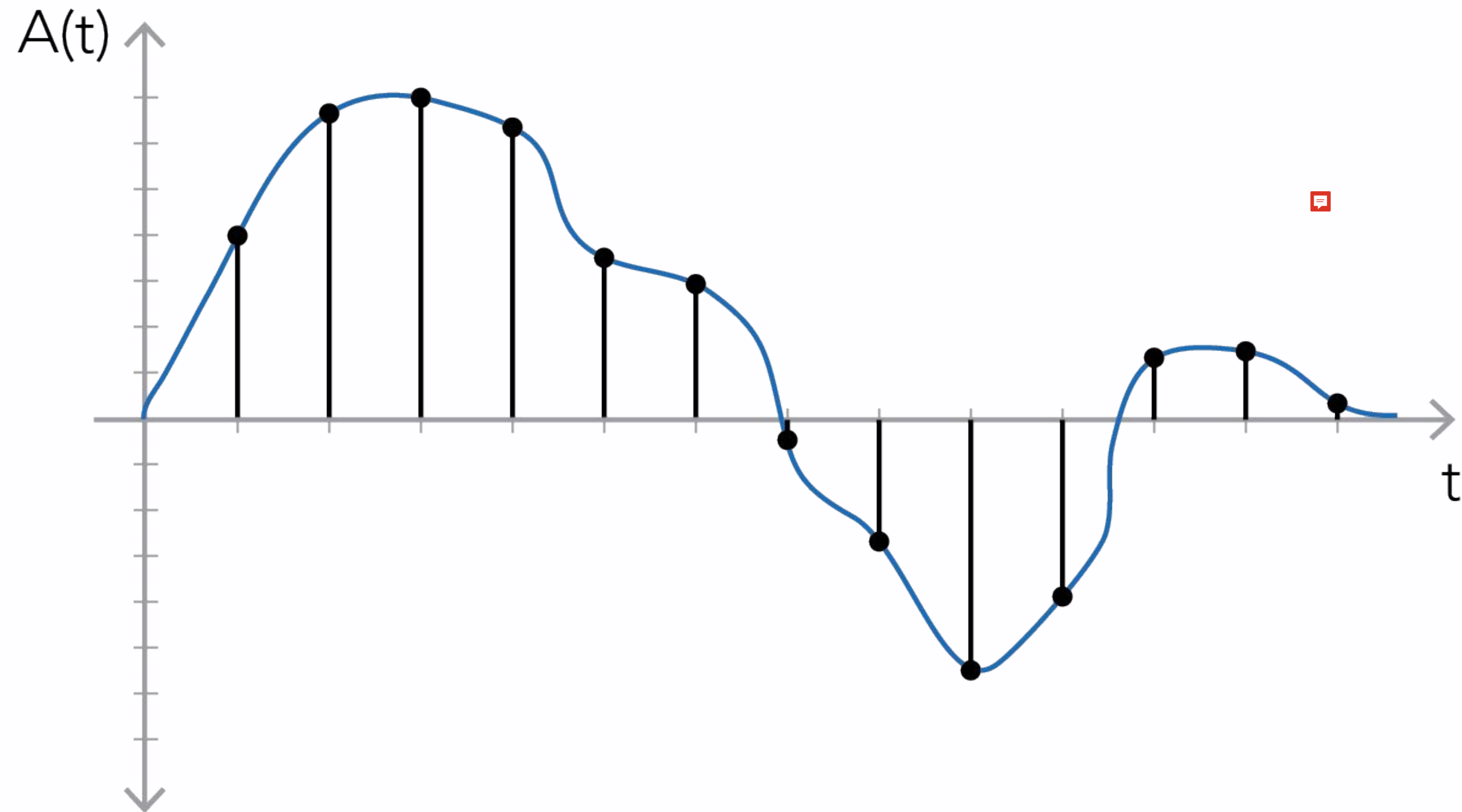
AD WANDLUNG?

	ZEIT (X-ACHSE)	WERT (Y-ACHSE)
ANALOG	KONTINUIERLICH	KONTINUIERLICH
DISKRET	(ZEIT-)DISKRET	KONTINUIERLICH
DIGITAL	DISKRET	DISKRET

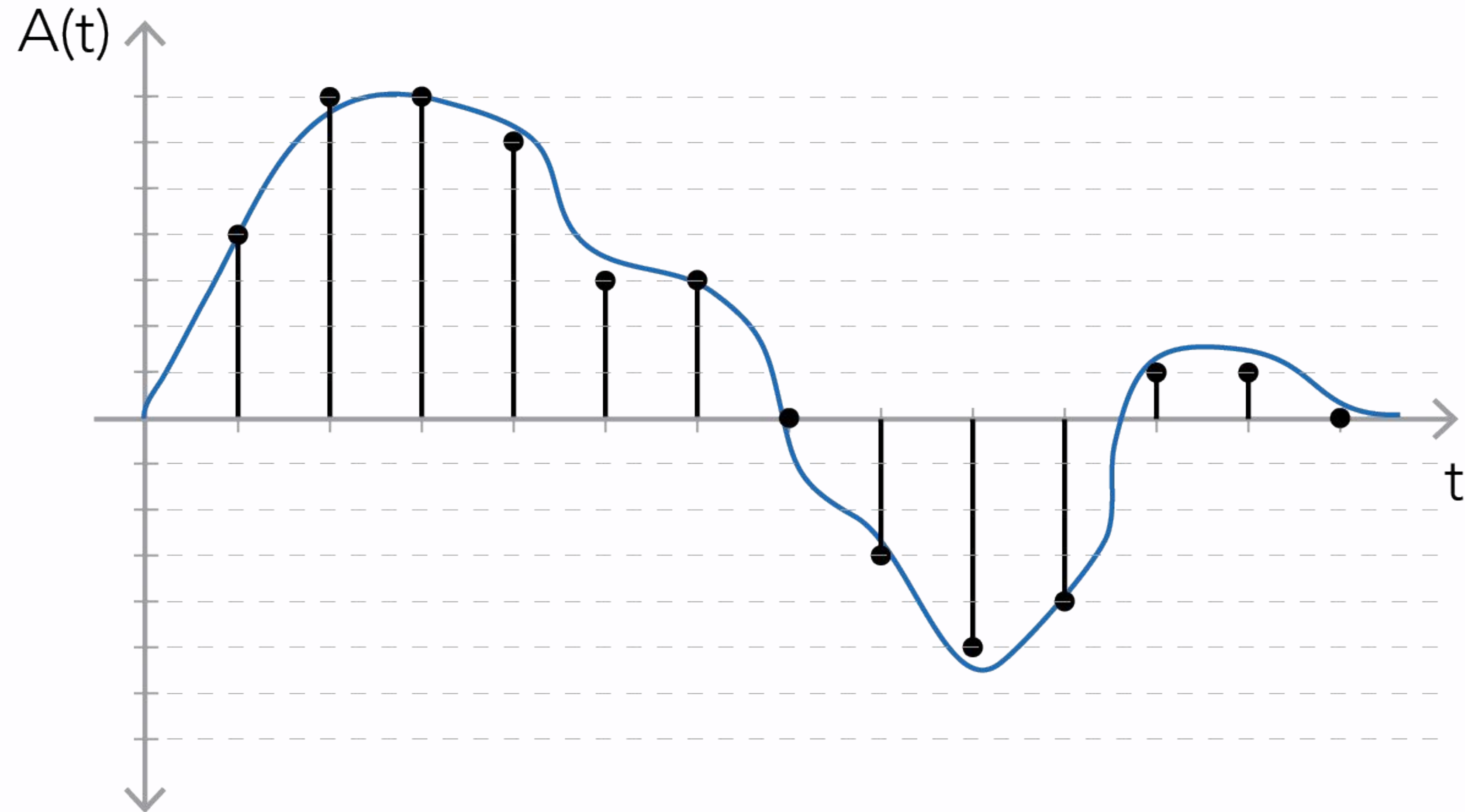
BEGRIFFE



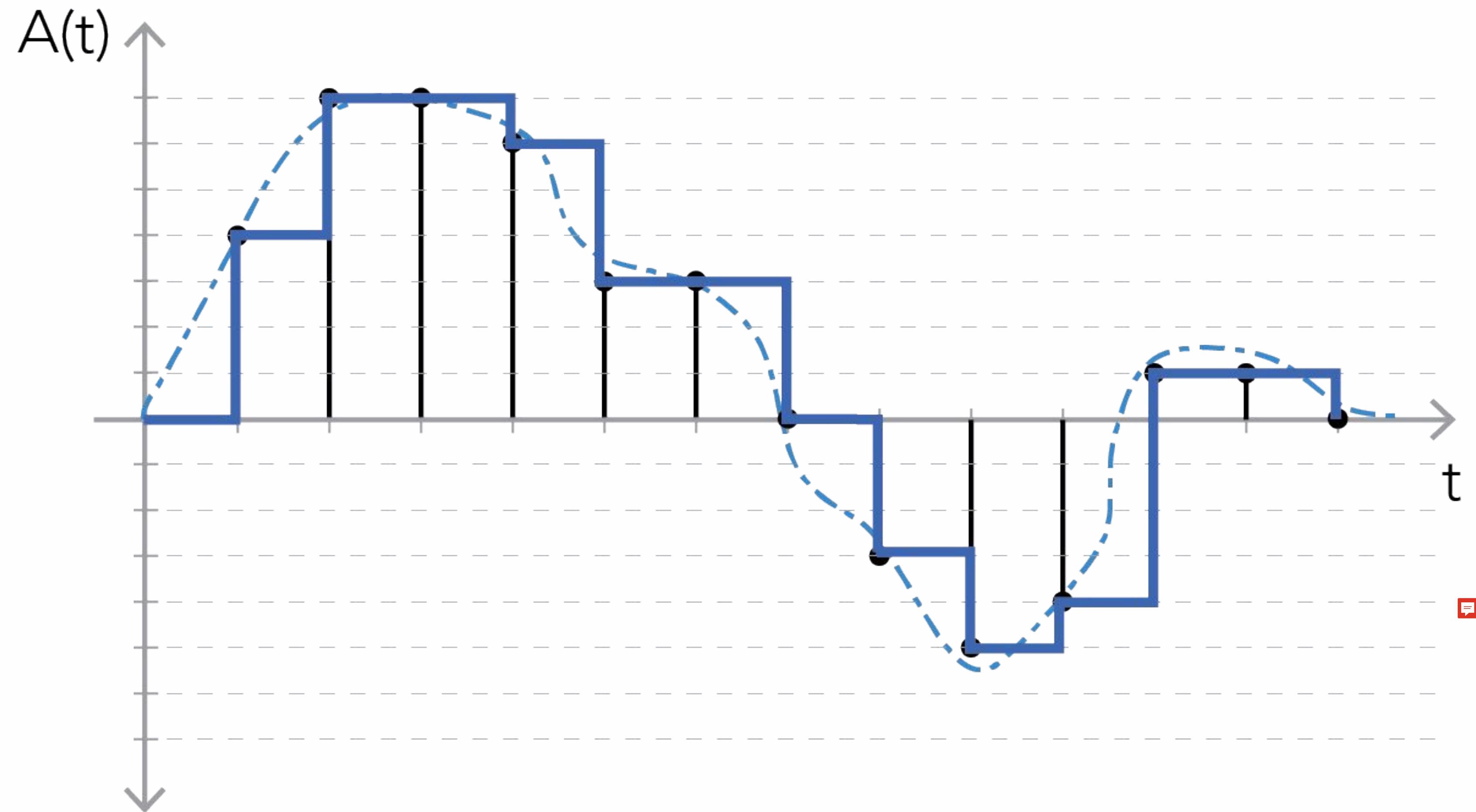
AD1 ANALOG



AD2 ZEITDISKRET



AD3 WERTDISKRET



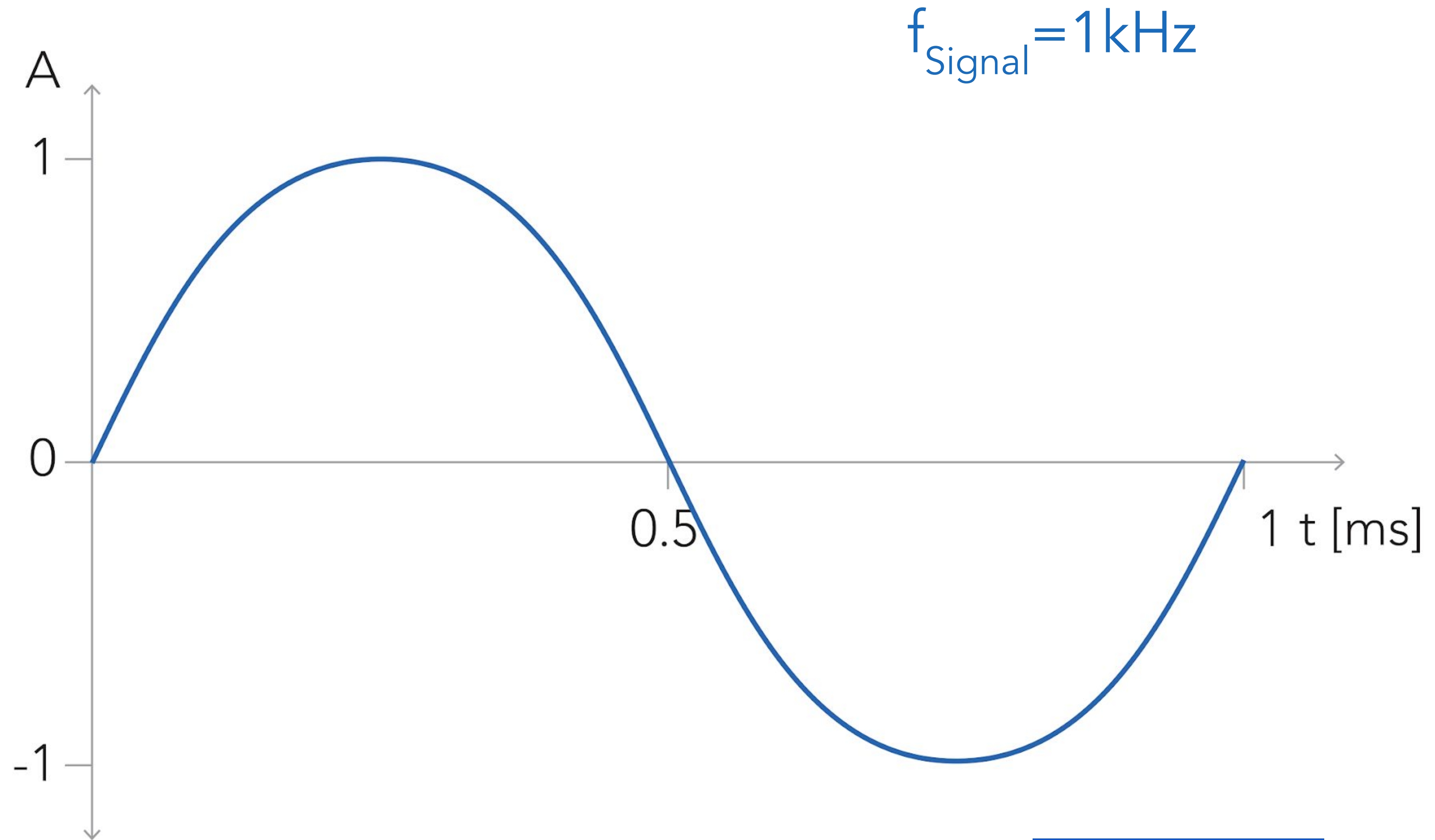
AD3 DIGITAL

$$f_{\text{Sampling}} > 2 \cdot f_{\text{Signal_Max}}$$

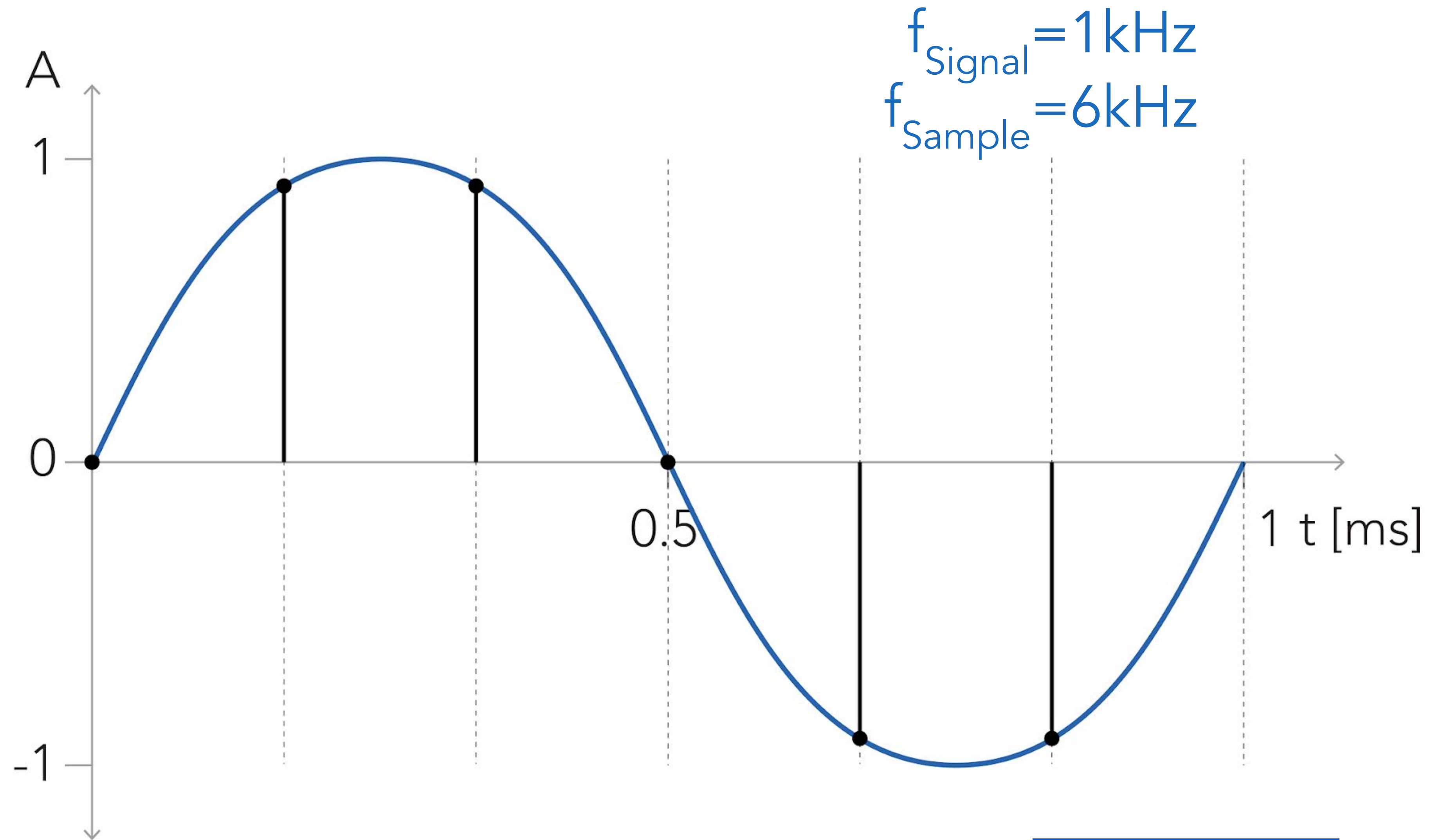
Nyquist/Shannon

Samplingfrequenz muss mind. das Doppelte der höchsten vorkommenden Signalfrequenz betragen

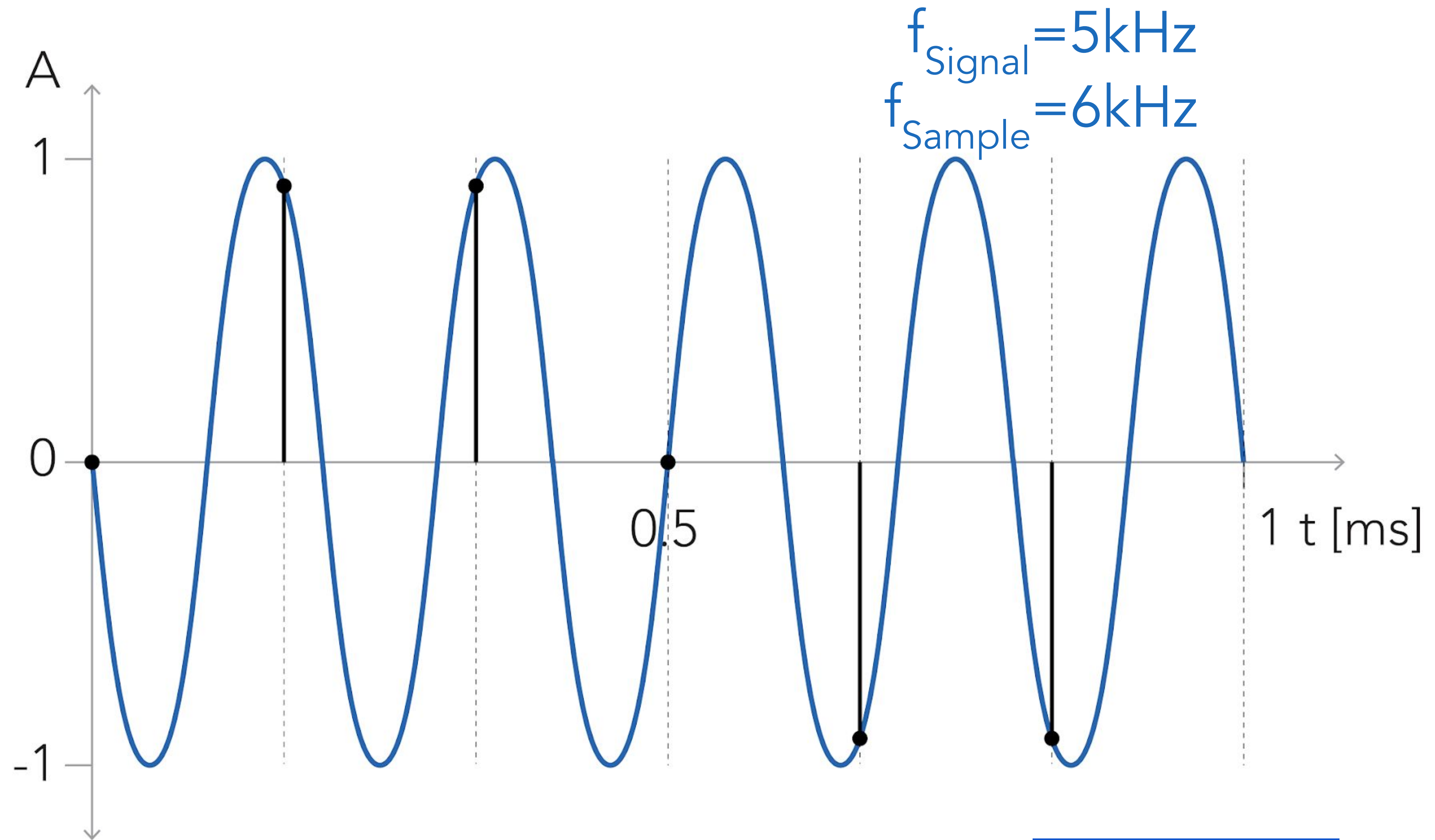
ABTASTTHEOREM



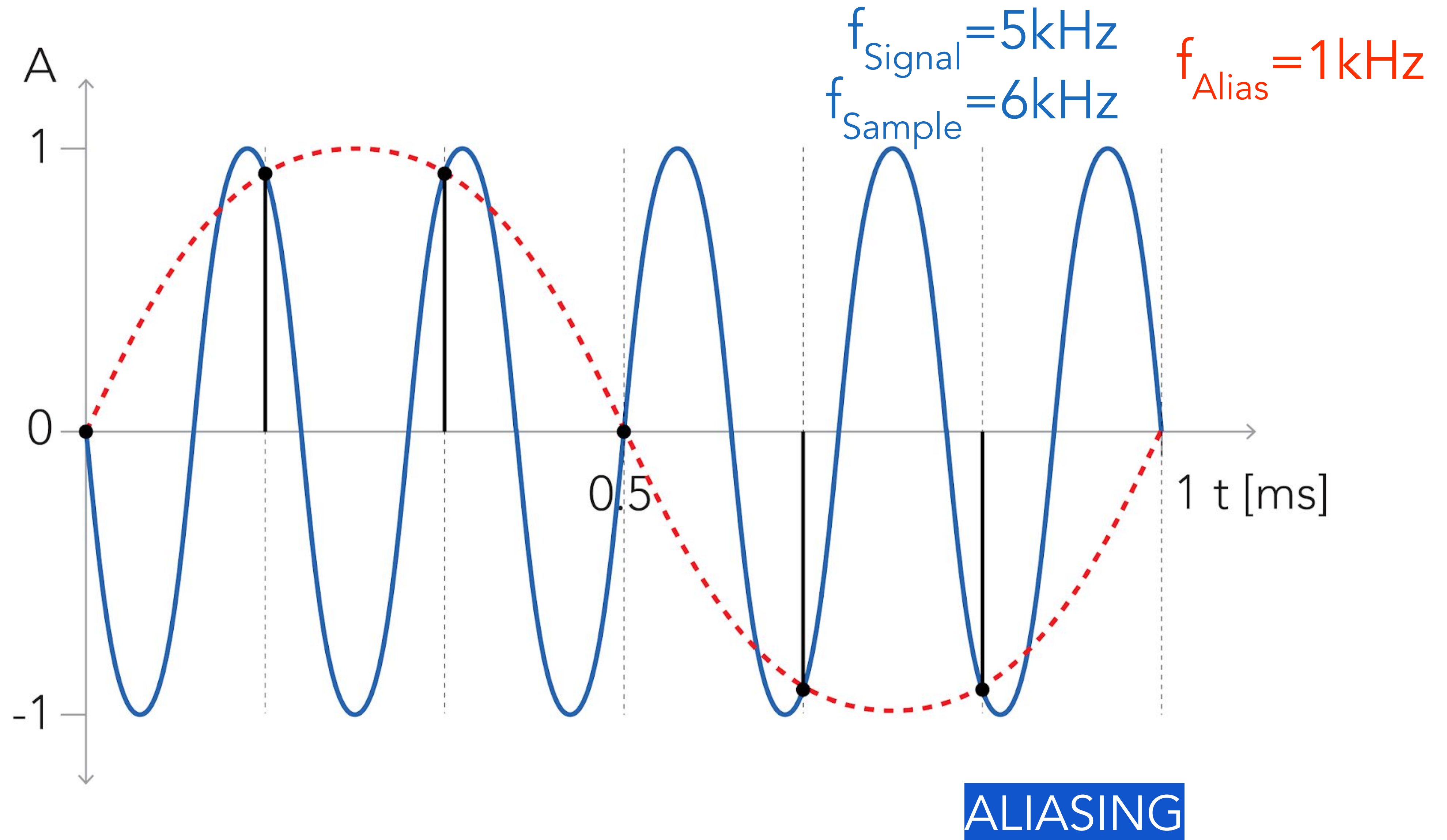
1KHZ SINUS

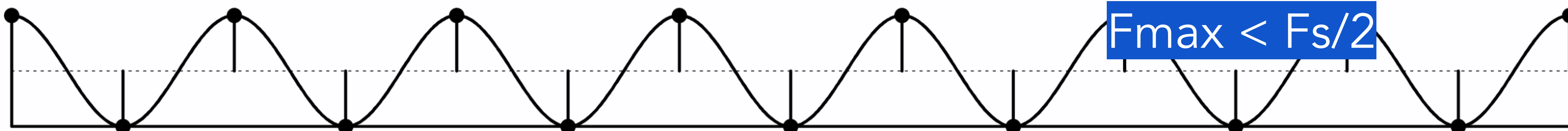
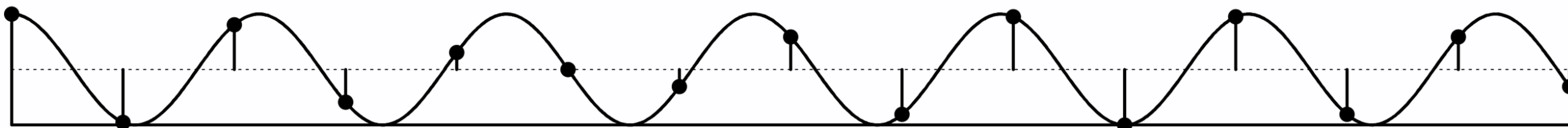
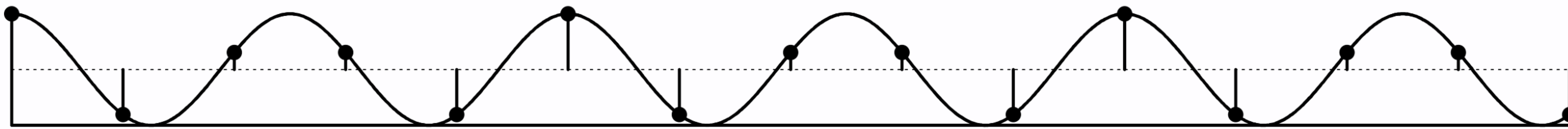
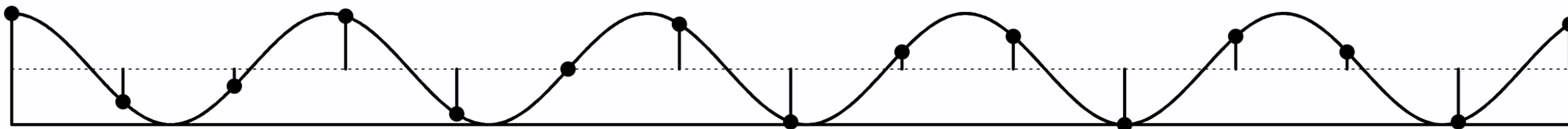
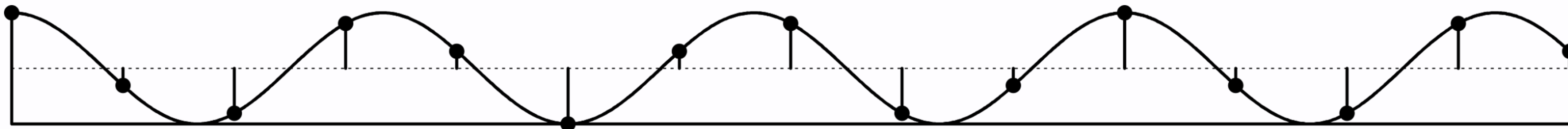


ABTASTUNG

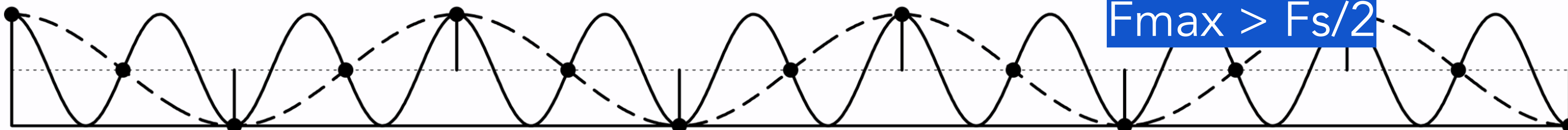
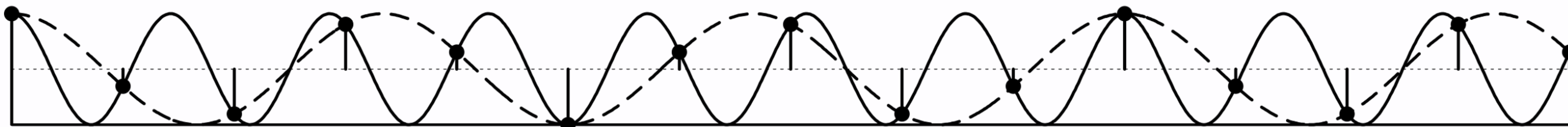
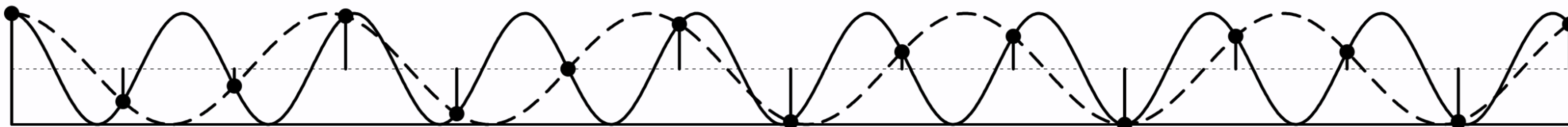
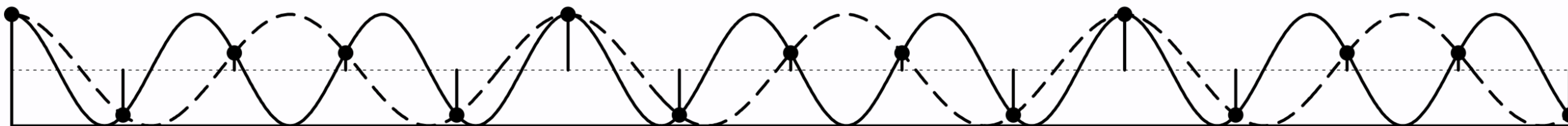
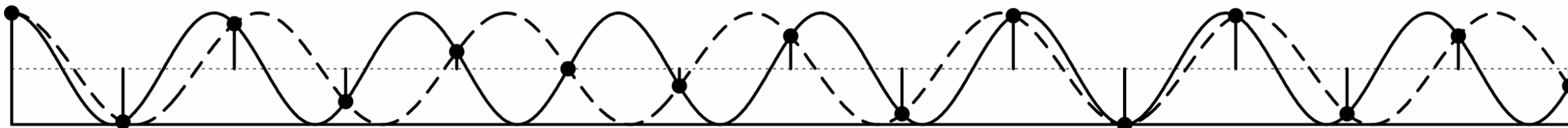


ABTASTUNG





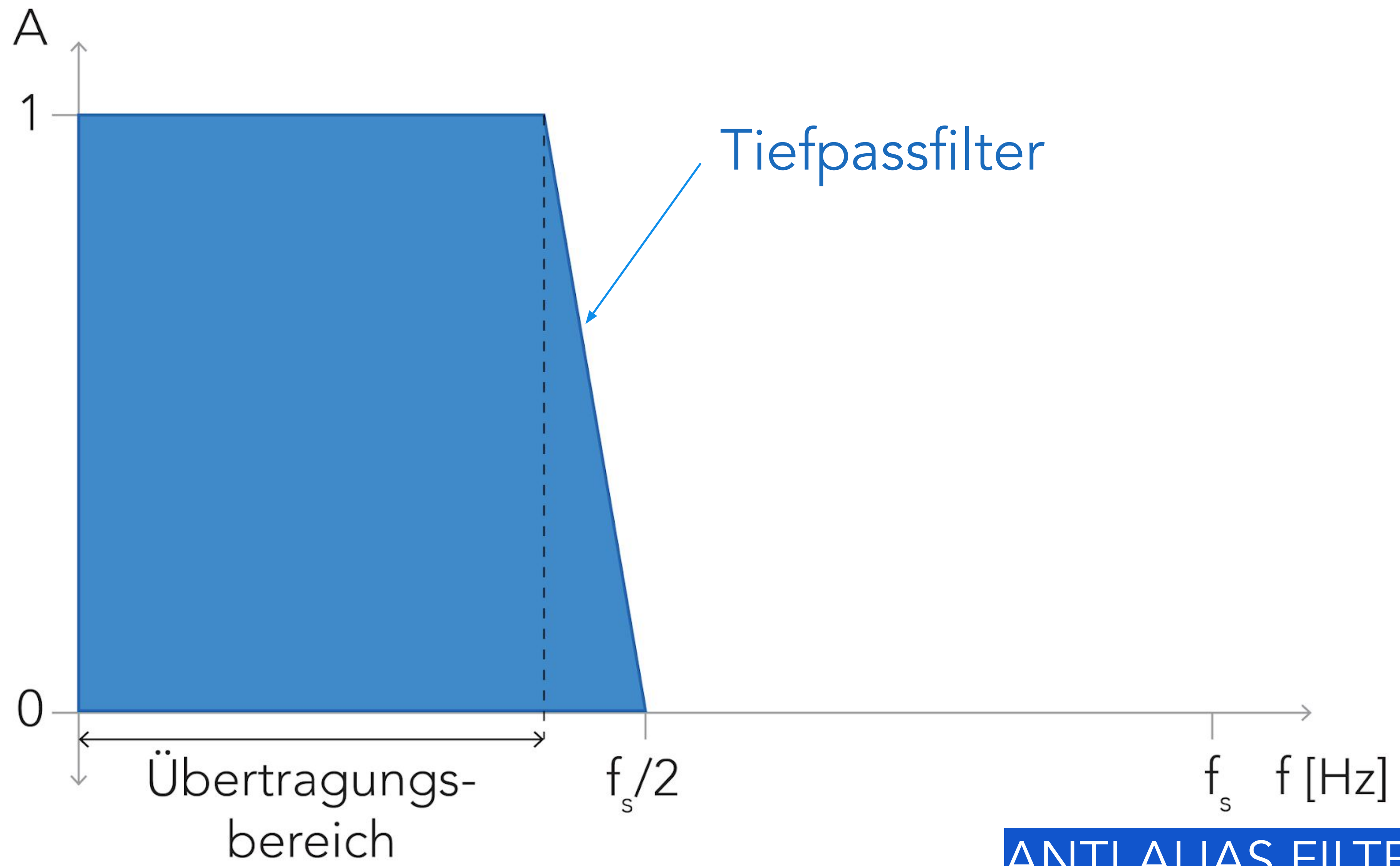
$F_{\max} < F_s/2$



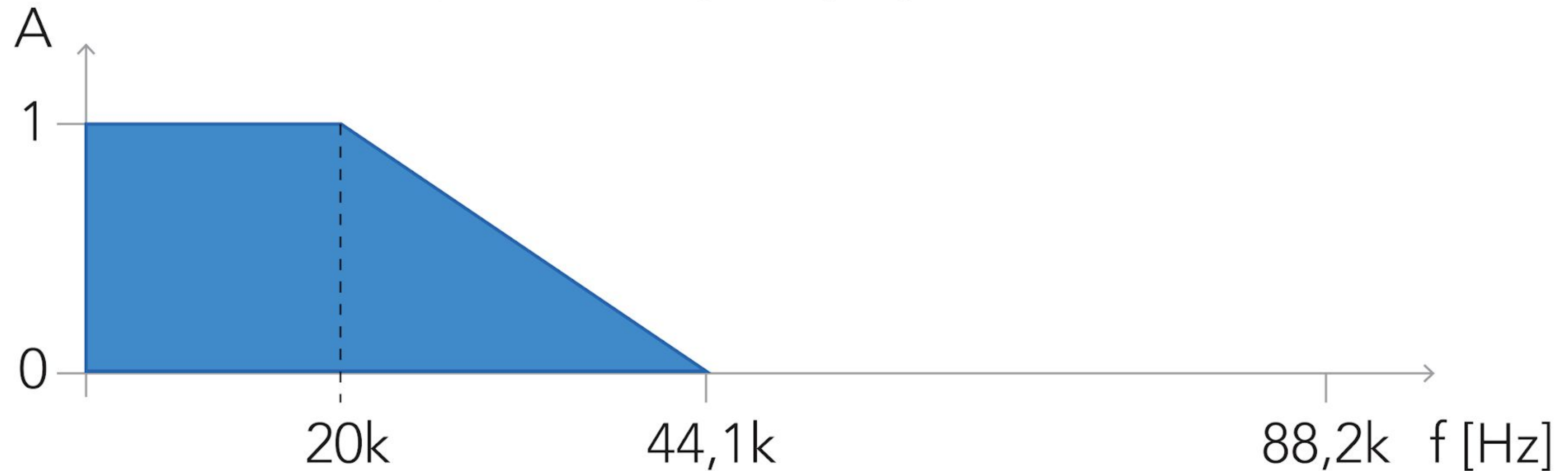
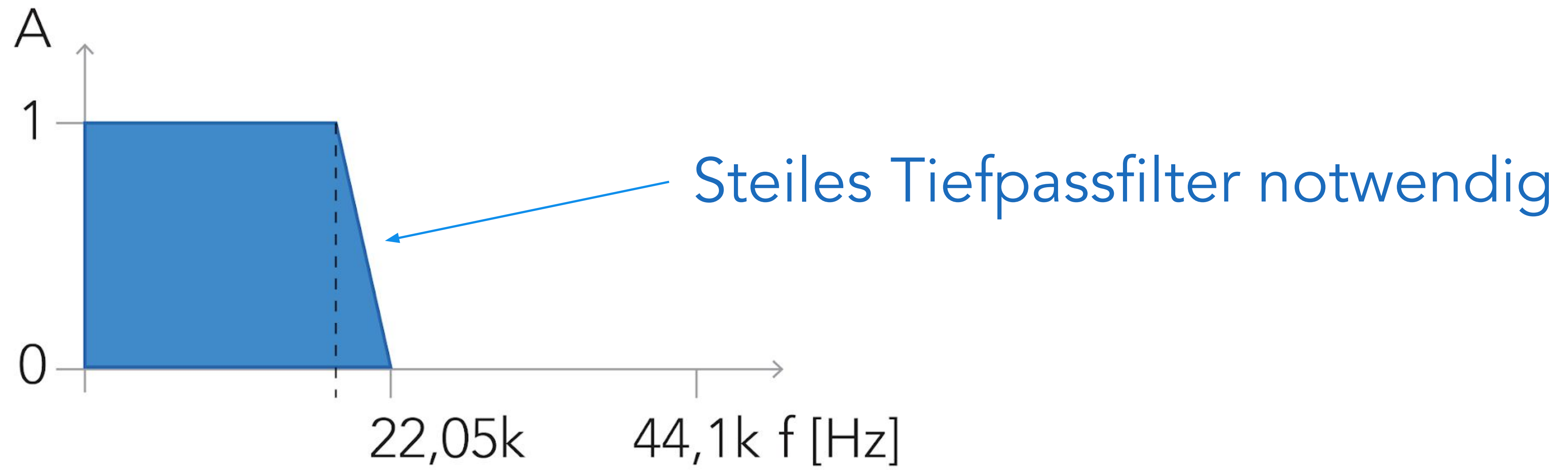
Hörbereich $< \sim 20\text{kHz}$
Samplerate $> 40\text{kHz}$

Typ. Digitalmedien
Audio CD 44,1kHz
Video 48kHz

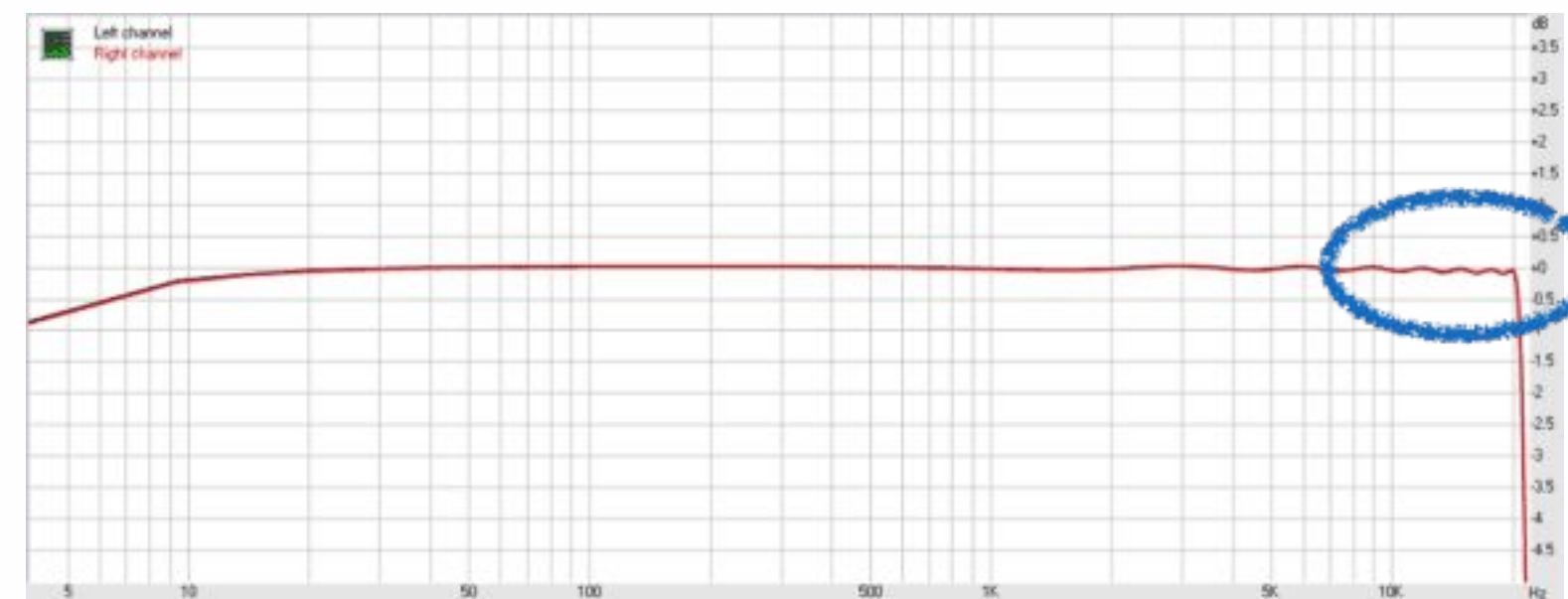
ABTASTFREQUENZ
TONAUFNAHME



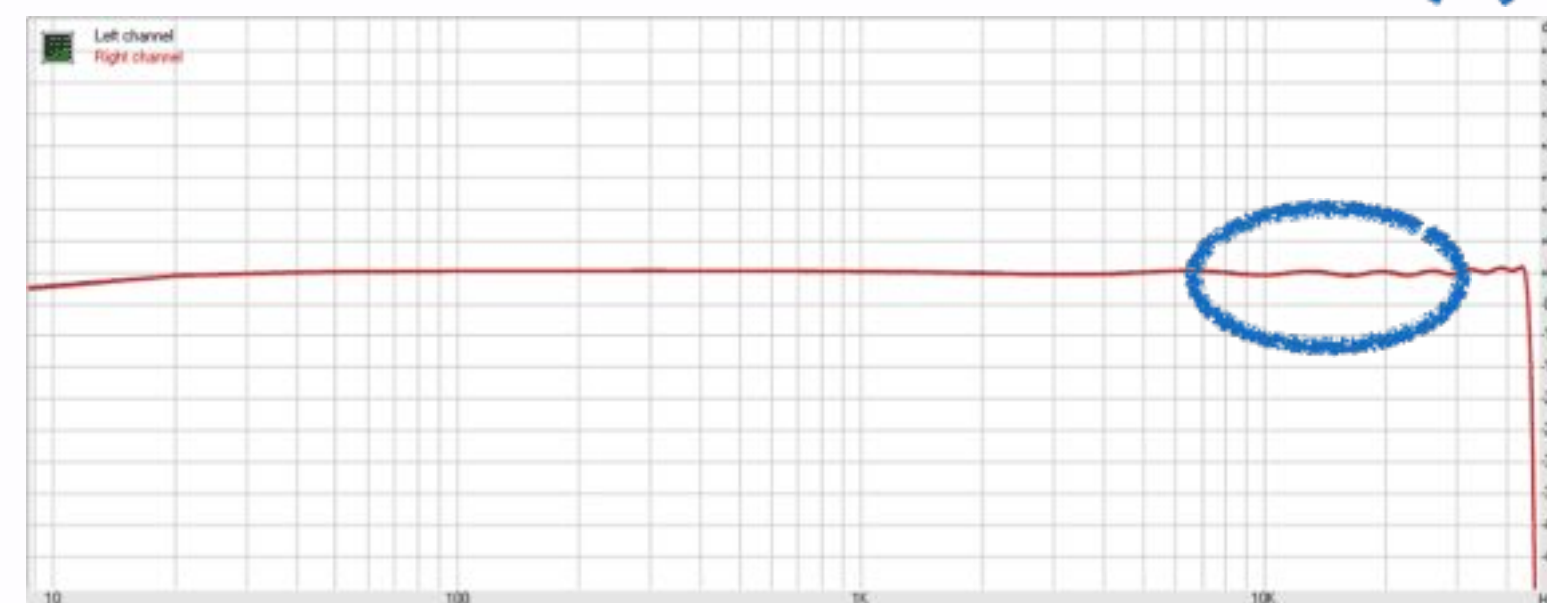
ANTI ALIAS FILTER



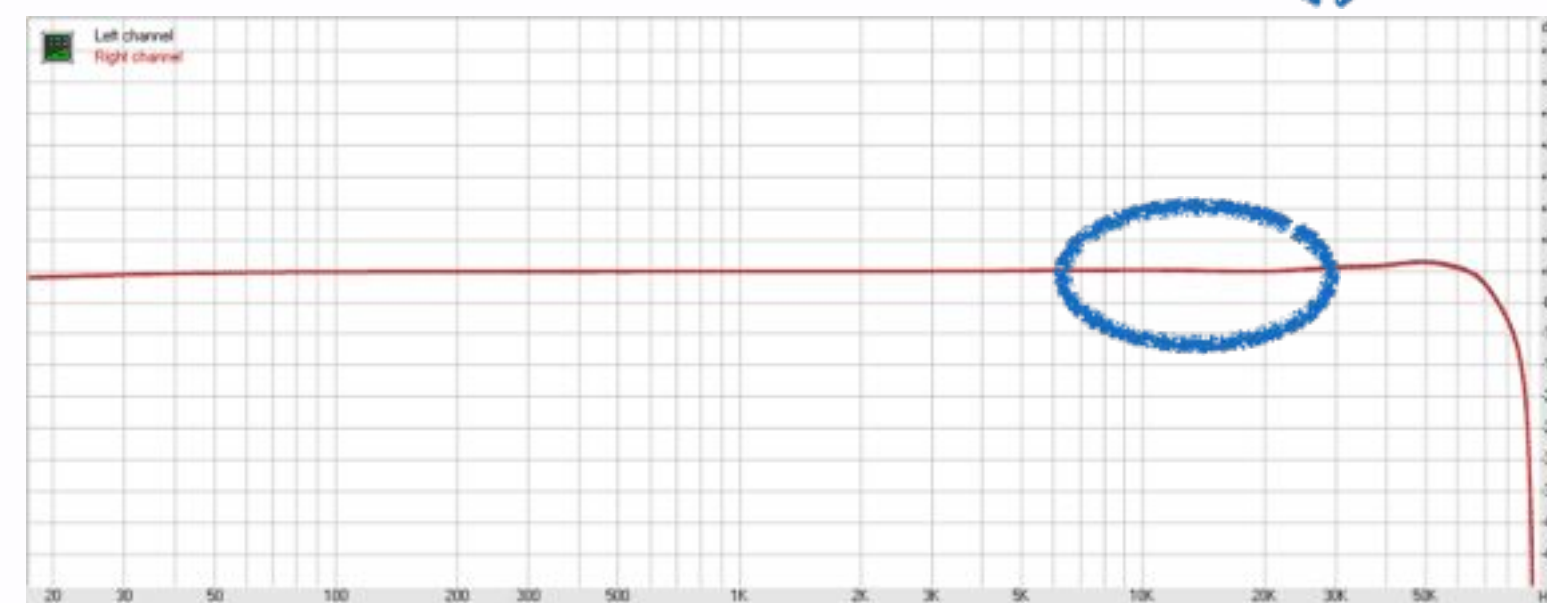
OVERSAMPLING



$F_s =$
44,1kHz



$F_s =$
96kHz



$F_s =$
192kHz

FREQUENZGANG
MOTU M4